

Exploratory Data Analysis for SPROJ

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PACKAGES

In order to do this Exploratory Data Analysis, I will use the following packages:

```
library(ggplot2) # for plotting data
library(readxl) # for reading excel files if necessary
library(dplyr) # for data wrangling
# library(arrow)
library(janitor) # for cleaning messy names of datasets
library(readr) # for reading files
library(gt) # for making 'publication ready' tables
library(usethis) #enables git pushing to put on github
library(gitcreds)
library(purrr) # this package is to loop and automate this entire process, instead of going t
library(fixest) #fixed fx
library(broom)
library(haven) ## allows to read .dta files meant for stata
library(tidyverse)
```

This senior project document will serve as the backbone of methodology and analysis part of the Senior Project”

Motivation

According to the UN Trade Hub and Development:

(<https://unctadstat.unctad.org/datacentre/reportInfo/US.RCA>)

“Revealed comparative advantage (RCA) is based on Ricardian trade theory, which posits that patterns of trade among countries are governed by their relative differences in productivity.

Although such productivity differences are difficult to observe, an RCA metric can be readily calculated using trade data to "reveal" such differences. While the metric can be used to provide a general indication and first approximation of a country's competitive export strengths, it should be noted that applied national measures which affect competitiveness such as tariffs, non-tariff measures, subsidies and others are not taken into account in the RCA metric."

"When a country has a revealed comparative advantage for a given product ($RCA > 1$), it is inferred to be a competitive producer and exporter of that product relative to a country producing and exporting that good at or below the world average. A country with a revealed comparative advantage in product i is considered to have an export strength in that product. The higher the value of a country's RCA for product i , the higher its export strength in product i ."

Part of this senior project aims to explain shifts in Revealed Comparative Advantage through a political economic lens, specifically in Mexico's Agriculture industry. As the UN mentions, this RCA is merely an inference variable, and is used as a 'first approximation' in comparative advantage. Many of the shifts in RCA over time are not captured. I will use classical economic theory, neoclassical theory, economic literature, and papers to describe shifts in the RCA through a political economic lens. My sector analysis in these documents will be done in order of sector code and number, starting from 01; Animals Live, throughout the rest of the agricultural sector.

The agricultural sector is one that is extremely broad- ranging from plants and livestock, to coffee beans and even honey. In order to further analyze the RCA index, we must discern what codes constitute the agriculture industry. Each sector has its own general product code at the 2 digit, 4 digit, and 6 digit level, with each digit being more specific in each product. Using the International Trade Administration (ITA) Harmonized System (HS) codes, we will use these definitions to discern what products are part of the agriculture industry. There is opinions in the literature that shows variations in the product codes that is part of the 'agricultural sector.' I will use the definitions from this paper (https://www.researchgate.net/publication/332551060_Changes_in_Foreign_Trade_in_Agricultural_Products) to classify the codes that belongs in the agricultural sector. Furthermore, I will analyze each sector at a general level, starting at the 2 digit, and then up to 4 digits. Due to constraints and limitations of the publicly available data, the maximum scope of analysis is up to the 4 digit HS code level. The data will also be split up into multiple time periods. Period A will be from 1992-2000, period B will be from 2000-2008, and period C will be from 2008-2016, period D will be from 2016-2024.

Functions

The section below describes the functions which are necessary in computing this analysis, I have made comments in the code chunks in order to further expand on each line of code and what it does.

Necessary Variables Function

```
## This is a function that takes in only the necessary variables for an RCA analysis, including  
  
necessary_variables <- function(df) { # inputs a dataframe  
  df |>  
    dplyr::mutate(cmdCode = as.character(cmdCode)) |>  
    arrange(period) |> #arranges the df by year  
    select(period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue) #selects  
}
```

RCA Function

```
## RCA calculation function  
  
#inputs country exports of product i , country's total exports, world exports of product i, and world total  
calculate_rca <- function(country_product, country_total, world_product, world_total) {  
  
  #join by period to align years  
  rca_df <- country_product |>  
    inner_join(country_total, by = "period", suffix = c("_prod", "_total")) |>  
    inner_join(world_product, by = "period") |>  
    inner_join(world_total, by = "period", suffix = c("_world_prod", "_world_total")) |>  
    mutate(  
      RCA = (fobvalue_prod / fobvalue_total) /  
            (fobvalue_world_prod / fobvalue_world_total)  
    ) |>  
    select(period, RCA)  
  
  return(rca_df)  
}
```

generalized RCA Function

```
calculate_rca_generalized <- function(country_sector_df,  
                                     world_sector_df,  
                                     country_all_df,  
                                     world_all_df) {
```

```

#first, I aggregate by year and product, with variables period, and cmdCode, removing any NA
country_prod <- country_sector_df |>
  group_by(period, cmdCode) |>
  summarise(country_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

world_prod <- world_sector_df |>
  group_by(period, cmdCode) |>
  summarise(world_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

#next, I aggregate total exports using fobvalue (denominator)
country_all <- country_all_df |>
  group_by(period) |>
  summarise(total_country_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

world_all <- world_all_df |>
  group_by(period) |>
  summarise(total_world_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

# 3. next, I merge everything by year and commodity code
rca_df <- country_prod |>
  left_join(world_prod, by = c("period", "cmdCode")) |>
  left_join(country_all, by = "period") |>
  left_join(world_all, by = "period") |>
  mutate(
    RCA = (country_product_value / total_country_value) /
      (world_product_value / total_world_value),
    has_CA = RCA >= 1
  ) |>
  arrange(period, cmdCode)

return(rca_df) #outputs a dataframe
}

```

Recorded loading dataframes

```

#setup

base_path <- "~/Desktop/SPROJ/workplace/input"

```

```

time_periods <- c("a", "b", "c")

sector_codes <- c("01", "02", "04", "05", "07", "08", "09", "10", "11",
                  "12", "13", "14", "15", "16", "17", "18", "19", "20",
                  "21", "22", "23", "24", "50", "51", "52", "53")

# functions (unchanged from your original)

necessary_variables <- function(df) {
  df |>
    dplyr::mutate(cmdCode = as.character(cmdCode)) |>
    arrange(period) |>
    select(period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue)
}

calculate_rca_generalized <- function(country_sector_df,
                                     world_sector_df,
                                     country_all_df,
                                     world_all_df) {
  country_prod <- country_sector_df |>
    group_by(period, cmdCode) |>
    summarise(country_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  world_prod <- world_sector_df |>
    group_by(period, cmdCode) |>
    summarise(world_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  country_all <- country_all_df |>
    group_by(period) |>
    summarise(total_country_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  world_all <- world_all_df |>
    group_by(period) |>
    summarise(total_world_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  rca_df <- country_prod |>
    left_join(world_prod, by = c("period", "cmdCode")) |>
    left_join(country_all, by = "period") |>
    left_join(world_all, by = "period") |>
    mutate(
      RCA = (country_product_value / total_country_value) /

```

```

        (world_product_value / total_world_value),
        has_CA = RCA >= 1
    ) |>
    arrange(period, cmdCode)

    return(rca_df)
}

#LOAD AND COMBINE TOTAL EXPORT FILES ACROSS TIME PERIODS

load_combined_totals <- function(folder_prefix, file_prefix, time_periods, base_path) {
  # Loops over time periods a, b, c and stacks into one dataframe
  map_dfr(time_periods, function(t) {
    file_path <- file.path(base_path,
                           paste0(folder_prefix, t), # i.e mx_total_xports_world_time_a
                           paste0(file_prefix, t, ".csv")) # e.x mx_xports_all_time_a.csv
    if (!file.exists(file_path)) {
      message("File not found, skipping: ", file_path)
      return(NULL)
    }
    read_csv(file_path, show_col_types = FALSE) |> necessary_variables()
  })
}

# Combined Mexico total exports: 1990-2012
mx_xports_all <- load_combined_totals(
  folder_prefix = "mx_total_xports_world_time_",
  file_prefix   = "mx_xports_all_time_",
  time_periods  = time_periods,
  base_path     = base_path
)

# Combined world total exports: 1990-2012
world_xports_all <- load_combined_totals(
  folder_prefix = "world_total_xports_time_",
  file_prefix   = "world_xports_all_time_",
  time_periods  = time_periods,
  base_path     = base_path
)

```

```

# PROCESS SECTOR ACROSS ALL THREE TIME PERIODS

process_sector <- function(sector) {
  sector_data <- map_dfr(time_periods, function(t) {

    # mx files: try short name first (time_a and time_c pattern)
    mx_file <- file.path(base_path,
                        paste0("mx_xports_prod_i_time_", t),
                        paste0("mx_sector_", sector, "_time_", t, ".csv"))

    # if not found, try long name (time_b pattern and sector 01)
    if (!file.exists(mx_file)) {
      mx_file <- file.path(base_path,
                          paste0("mx_xports_prod_i_time_", t),
                          paste0("mx_xports_sector_", sector, "_time_", t, ".csv"))
    }

    # world files: always world_xports_sector_XX_time_Y.csv
    world_file <- file.path(base_path,
                          paste0("world_xports_prod_i_time_", t),
                          paste0("world_xports_sector_", sector, "_time_", t, ".csv"))

    if (!file.exists(mx_file) | !file.exists(world_file)) {
      message("Skipping sector ", sector, " time_", t, " (file not found)")
      return(NULL)
    }

    mx_sector <- read_csv(mx_file, show_col_types = FALSE) |>
      necessary_variables()

    world_sector <- read_csv(world_file, show_col_types = FALSE) |> necessary_variables()

    rca_slice <- calculate_rca_generalized(
      mx_sector,
      world_sector,
      mx_xports_all,
      world_xports_all
    )

    return(rca_slice)
  })
}

```

```

    if (is.null(sector_data) | nrow(sector_data) == 0) return(NULL)
    sector_data$sector <- sector
    return(sector_data)
}

# BUILD THE BIG RCA DATAFRAME

dir.create("output/clean/rca_sectors", recursive = TRUE, showWarnings = FALSE)

rca_big_df <- map_dfr(sector_codes, function(s) {
  result <- process_sector(s)
  if (!is.null(result)) {
    write_csv(result, paste0("output/clean/rca_sectors/rca_sector_", s, ".csv"))
    message("Saved sector ", s)
  }
  return(result)
})

# Add sector column and NAFTA dummies
rca_big_df <- rca_big_df |>
  mutate(
    cmdCode = as.character(cmdCode),
    sector = substr(cmdCode, 1, 2),
    post_nafta = if_else(period >= 1994, 1, 0),
    nafta_phase = case_when(
      period < 1994 ~ "pre",
      period >= 1994 & period <= 2000 ~ "post"
    )
  )
)

```

SECTORS

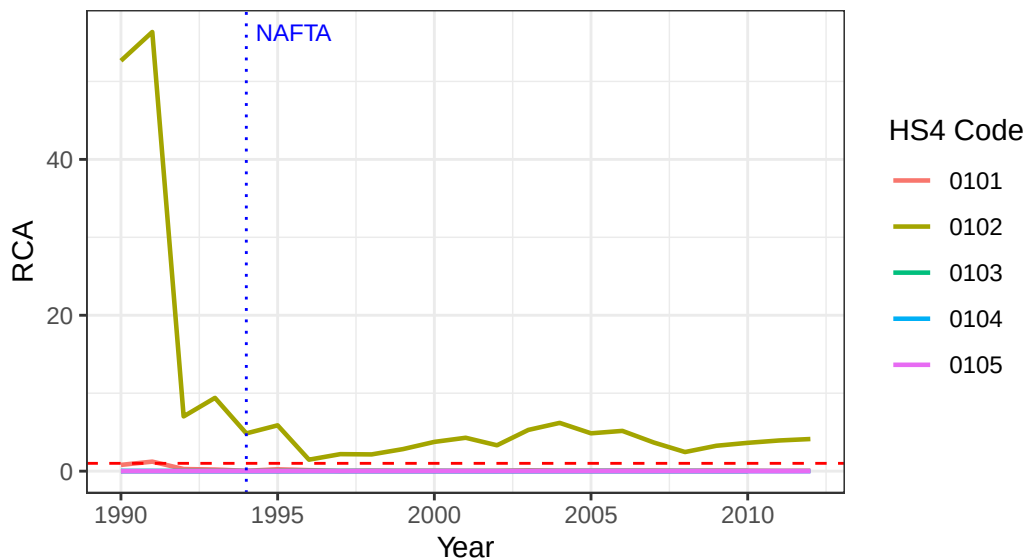
01 Animals; Live

This section will calculate RCA over time before NAFTA of the animal section, containing HS product code of 0101 to 0106.


```
## load the datasets
rca_sector_01 <- rca_big_df |>
  filter(sector == "01", nchar(cmdCode) == 4)

ggplot(rca_sector_01, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") + # NAFTA line
  annotate("text", x = 1994.3, y = max(rca_sector_01$RCA, na.rm = TRUE),
    label = "NAFTA", hjust = 0, size = 3, color = "blue") +
  labs(title = "RCA Over Time - Sector 01: Animals Live",
    subtitle = "1990-2012",
    x = "Year", y = "RCA",
    color = "HS4 Code") +
  theme_bw()
```

RCA Over Time – Sector 01: Animals Live
1990–2012

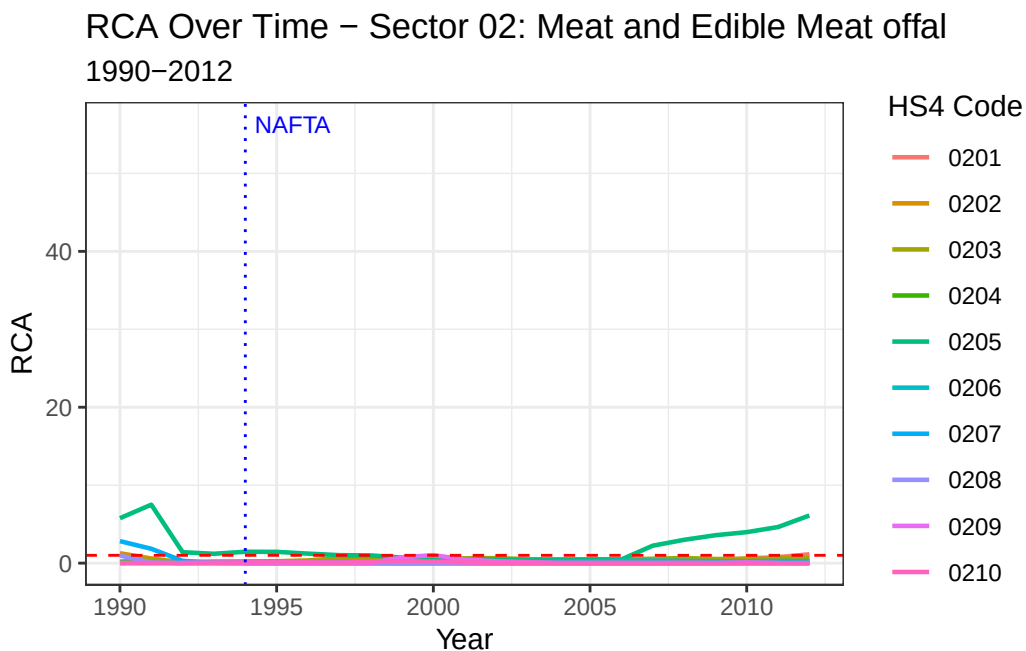


Interesting that Mexico seems to have a general RCA of Live Animals, but there was a sharp decline starting in 1993. NAFTA was enacted January 1, 1994- was it a driving factor in RCA shift? What about data from future years?

02 Meat and Edible Meat offal

```
## load the datasets
rca_sector_02 <- rca_big_df |>
  filter(sector == "02", nchar(cmdCode) == 4)

## graph RCA
ggplot(rca_sector_02, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") + # NAFTA line
  annotate("text", x = 1994.3, y = max(rca_sector_01$RCA, na.rm = TRUE),
    label = "NAFTA", hjust = 0, size = 3, color = "blue") +
  labs(title = "RCA Over Time - Sector 02: Meat and Edible Meat offal",
    subtitle = "1990-2012",
    x = "Year", y = "RCA",
    color = "HS4 Code") +
  theme_bw()
```



Sector of 02 seems to not have much of a comparative advantage in Mexico.

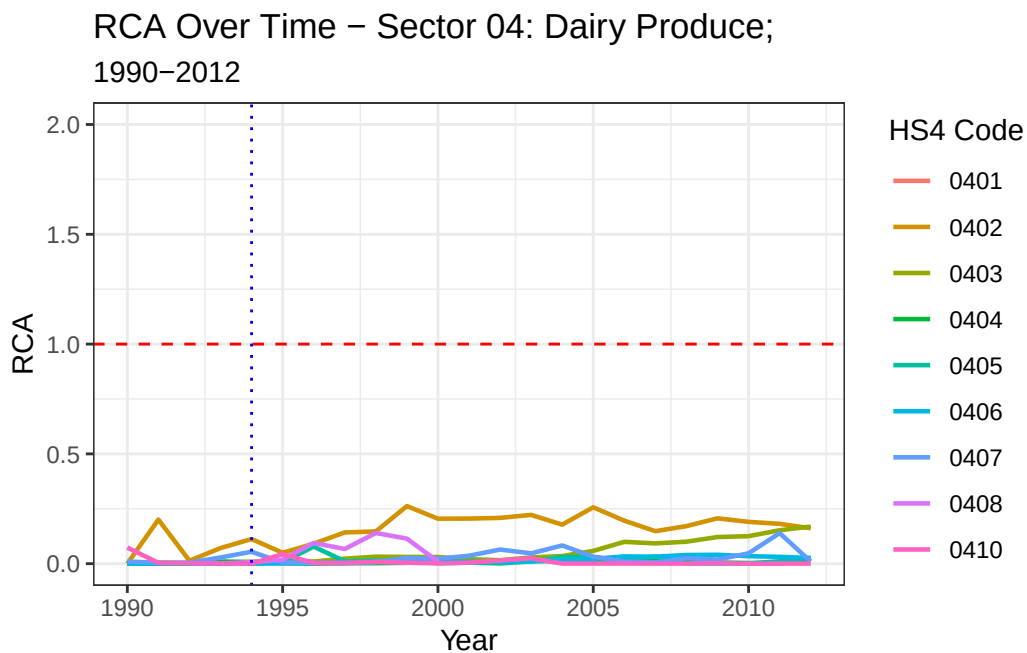
04 Dairy Produce; Birds' eggs; Natural Honey; Edible products of animal origin

- Omit 0409: Honey.

```
rca_sector_04 <- rca_big_df |>
  filter(sector == "04", nchar(cmdCode) == 4)
```

Graph Sector 04

```
## graph RCA
ggplot(rca_sector_04, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 2)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 04: Dairy Produce;",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



05 Other Animal Products

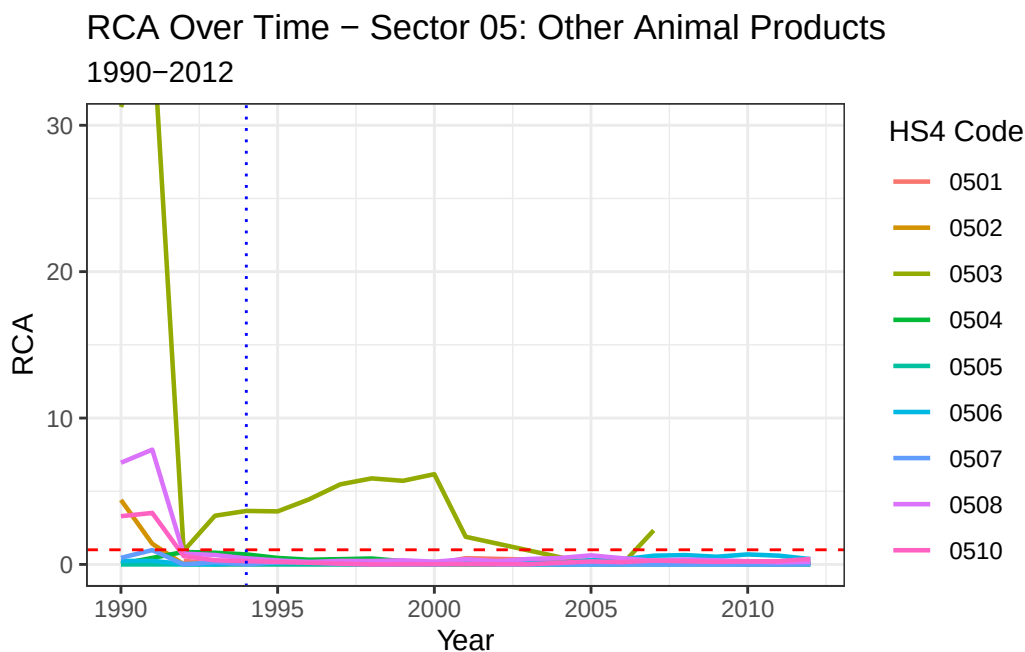
ANIMAL ORIGINATED PRODUCTS; NOT ELSEWHERE SPECIFIED OR INCLUDED

We omit the following

- 0501 Human hair; unworked, whether or not washed or scoured; waste of human hair
- 0509 Sponges
- 0511 Animal products not elsewhere specified or included; dead animals of chapter 1 or 3, unfit for human consumption

```
rca_sector_05 <- rca_big_df |>
  filter(sector == "05", nchar(cmdCode) == 4)
```

```
ggplot(rca_sector_05, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 30)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 05: Other Animal Products",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

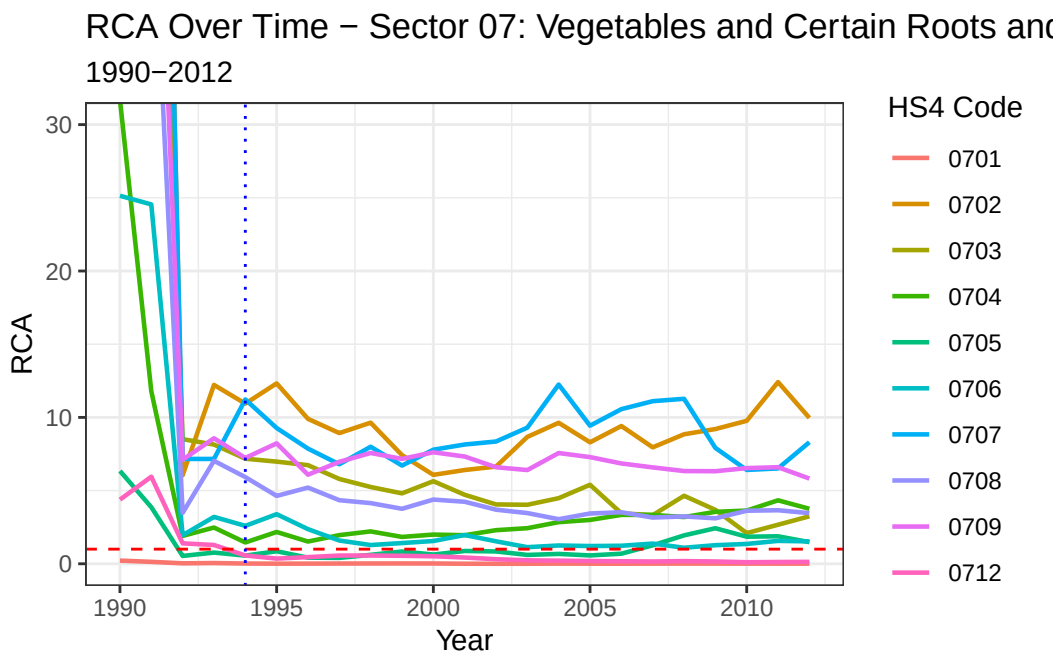


Interestingly, 0503 “Horsehair, waste” has a comparative advantage in that product, but the sector does not have a comparative advantage.

07- Vegetables and Certain Roots and Tubes; Edible

```
rca_sector_07 <- rca_big_df |>
  filter(sector == "07", nchar(cmdCode) == 4)
```

```
ggplot(rca_sector_07, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 30)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 07: Vegetables and Certain Roots and Tubes; Edible",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



Sector Analysis : 07

Now, I am tasked to clean the dataset and calculate an RCA given the dataset with the entire sector. Using the code above, find a way to generalize the RCA calculation to continue the calculation, with an ultimate goal of showing an RCA graph for each product in the sector.

Now, I must use the necessary variables for the RCA. necessary variables are the following: period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. Recall that the RCA variable is calculated with the following:

$RCA = [(Country \text{ Exports of Product } i / Country \text{ A Exports of ALL products}) / (World \text{ Exports of product } i / World \text{ Exports of all products})]$

$$RCA_i = \frac{\frac{X_{A,i}}{X_{A,all}}}{\frac{X_{World,i}}{X_{World,all}}}$$

The RCA variable calculates a share of country exports a product over the world exports of said product. If the RCA variable is greater than, or equal to 1, then the country has a revealed comparative advantage in producing said good. According to neoclassical theory, gains from trade are still probable in comparative advantage. Theory says to export goods in which the country has a comparative advantage in producing.

Now, I have the necessary variables, Country MX Exports of product 0701 -> 0710. I have the data of Mexico's exports of all products, titled `mx_xports_all`. Now, I need World's exports of products 0701 -> 0710. I can do this by using UNCOMTRADE data as Ill. Repeating the same steps, I will grab the necessary variables.

I now have World exports of product 0701 -> 0710 with the same time period of 1992 to 2000. Recall I have datasets of Mexico's exports of all products titled `mx_xports_all` with variables `period`, `reporterISO`, `cmdCode`, `flowCode`, `qty`, `netWgt`, `fobvalue`, `primaryValue`. I also have `world_xports_all` which contains world exports of all products with variables `period`, `reporterISO`, `cmdCode`, `flowCode`, `qty`, `netWgt`, `fobvalue`, `primaryValue`. HoIver, notice that `world_xports_sector_07` and `mx_xports_sector_07` have the `cmdCode`, hoIver, they are not aggregated, and their `cmdCode` varies from 0701 -> 0710. Below is an attempted geenralized calclation of RCA, with a clear difference of grouping by `group_by(period, cmdCode)` due to the nature of the dataframe with varying `cmdCode`

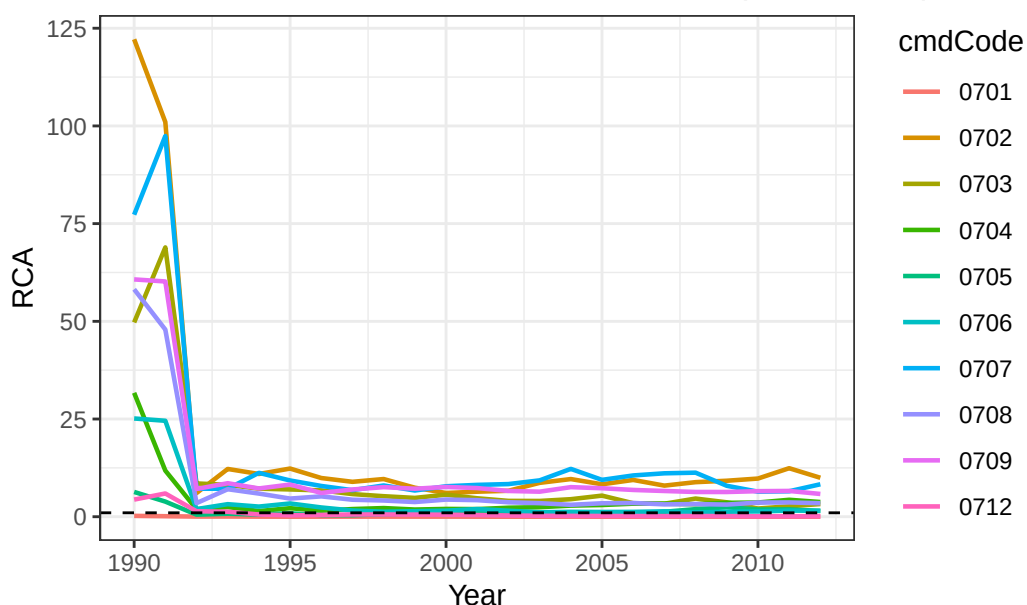
Let's test this in action to see if it works!

RCA of sector 0701 to 0710.

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_07, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (0701-0710)",
       x = "Year", y = "RCA") +
  theme_bw()
```

RCA Over Time – HS07 Sector Products (0701–0710)



I can now develop code to show which products have an $RCA > 1$ and $RCA < 1$.

I can do this by running a simple filter:

```
## show which years had an RCA > 1
table_RCA_gt1 <- rca_sector_07 |>
  filter(RCA > 1) |>
  arrange(period, cmdCode)
table_RCA_gt1
```

A tibble: 173 x 11

	period	cmdCode	country_product_value	world_product_value	total_country_value
	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	1990	0702	428402080	732965989	26344665088
2	1990	0703	80259696	337482057	26344665088
3	1990	0704	13531930	89220241	26344665088
4	1990	0705	3586518	118230254	26344665088
5	1990	0706	6576971	54681540	26344665088
6	1990	0707	79643672	215427053	26344665088
7	1990	0708	18968764	68132928	26344665088
8	1990	0709	245167568	843898327	26344665088
9	1990	0712	5677731	270895411	26344665088
10	1991	0702	261739168	736978466	26956705792

i 163 more rows

[
Mexico's RCA, Sector 07 (RCA > 1)] Mexico's RCA, Sector 07 (RCA > 1)

cmdCode

0701	
0702	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005
0703	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005
0704	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005
0705	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005
0706	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005
0707	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005
0708	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005
0709	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005
0712	

```
# i 6 more variables: total_world_value <dbl>, RCA <dbl>, has_CA <lgl>,  
#   sector <chr>, post_nafta <dbl>, nafta_phase <chr>
```

or by showing which years had an RCA > 1

```
#summarize the comparative advantage by product by grouping by product code  
summary_CA_by_product <- rca_sector_07 |>  
  group_by(cmdCode) |>  
  summarise(  
    years_with_CA = paste(period[RCA > 1], collapse = ", "),  
    n_years_CA = sum(RCA > 1),  
    total_years = n(),  
    share_years_CA = n_years_CA / total_years  
  )
```

and a publication ready table:

```
summary_CA_by_product |>  
  gt() |>  
  tab_header(  
    title = "Mexico's RCA, Sector 07 (RCA > 1)",  
  )
```

I can fix this dataframe by changing the name of the previous one

Sector definitions

07 - VEGETABLES AND CERTAIN ROOTS AND TUBERS; EDIBLE

- 0701 - Potatoes; fresh or chilled
- 0702 - Tomatoes; fresh or chilled
- 0703 - Onions, shallots, garlic, leeks and other alliacious vegetables; fresh or chilled
- 0704 - Cabbages, caulifloIrs, kohlrabi, kale and similar edible brassicas; fresh or chilled
- 0705 - Lettuce (*lactuca sativa*) and chicory (*cichorium spp.*) fresh or chilled
- 0706 - Carrots, turnips, salad beetroot, salsify, celeriac, radishes and similar edible roots; fresh or chilled
- 0707 - Cucumbers and gherkins; fresh or chilled
- 0708 - Leguminous vegetables; shelled or unshelled, fresh or chilled
- 0709 - Vegetables; n.e.c. in chapter 07, fresh or chilled
- 0710 - Vegetables (uncooked or cooked by steaming or boiling in water); frozen
- 0711 - Vegetables provisionally preserved; (e.g. by sulphur dioxide gas, in brine, in sulphur water or in other preservative solutions), but unsuitable in that state for immediate consumption
- 0712 - Vegetables, dried; whole, cut, sliced, broken or in powder, but not further prepared
- 0713 - Vegetables, leguminous; shelled, whether or not skinned or split, dried

08- FRUIT AND NUTS, EDIBLE; PEEL OF CITRUS FRUIT OR MELONS

We omit the following:

- 0812 Fruit and nuts provisionally preserved; e.g. by sulphur dioxide gas, brine, in sulphur water or in other preservative solutions, but unsuitable in that state for immediate consumption
- 0813 Fruit, dried, other than that of heading no. 0801 to 0806; mixtures of nuts or dried fruits of this chapter
- 0814 Peel of citrus fruit or melons (including watermelons); fresh, frozen dried or provisionally preserved in brine, in sulphur water or in other preservative solutions

```
# label: debug-check  
exists("rca_big_df")
```

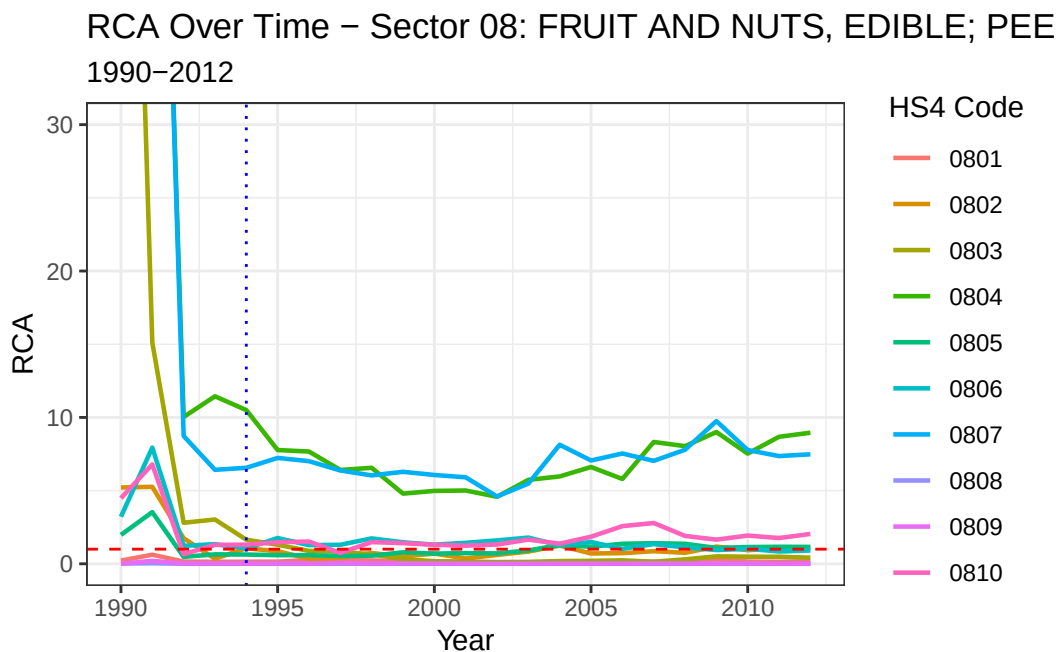
```
[1] TRUE
```

```
nrow(rca_big_df)
```

```
[1] 4305
```

```
rca_sector_08 <- rca_big_df |>  
  filter(sector == "08", nchar(cmdCode) == 4)
```

```
ggplot(rca_sector_08, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +  
  geom_line(size = 0.8) +  
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +  
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +  
  coord_cartesian(ylim = c(0, 30)) + # cap at 10, adjust as needed  
  labs(title = "RCA Over Time - Sector 08: FRUIT AND NUTS, EDIBLE; PEEL OF CITRUS FRUIT OR M",  
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +  
  theme_bw()
```



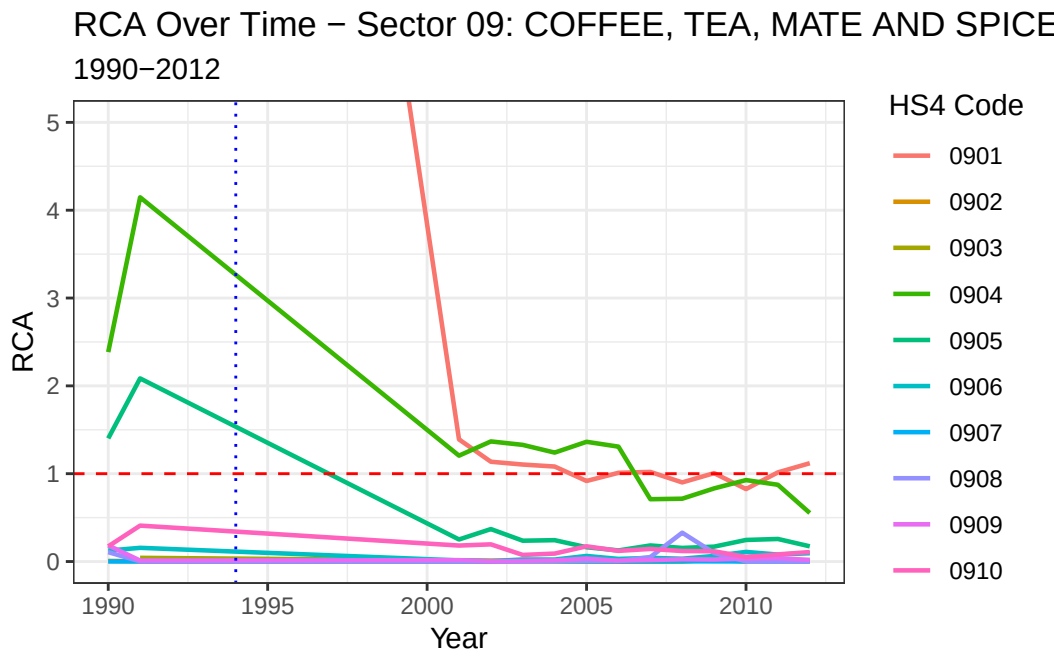
09- COFFEE, TEA, MATE AND SPICES

No 4 digit HS classification code is omitted.

```
rca_sector_09 <- rca_big_df |>
  filter(sector == "09", nchar(cmdCode) == 4)
```

Plot sector 09

```
ggplot(rca_sector_09, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 09: COFFEE, TEA, MATE AND SPICES",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



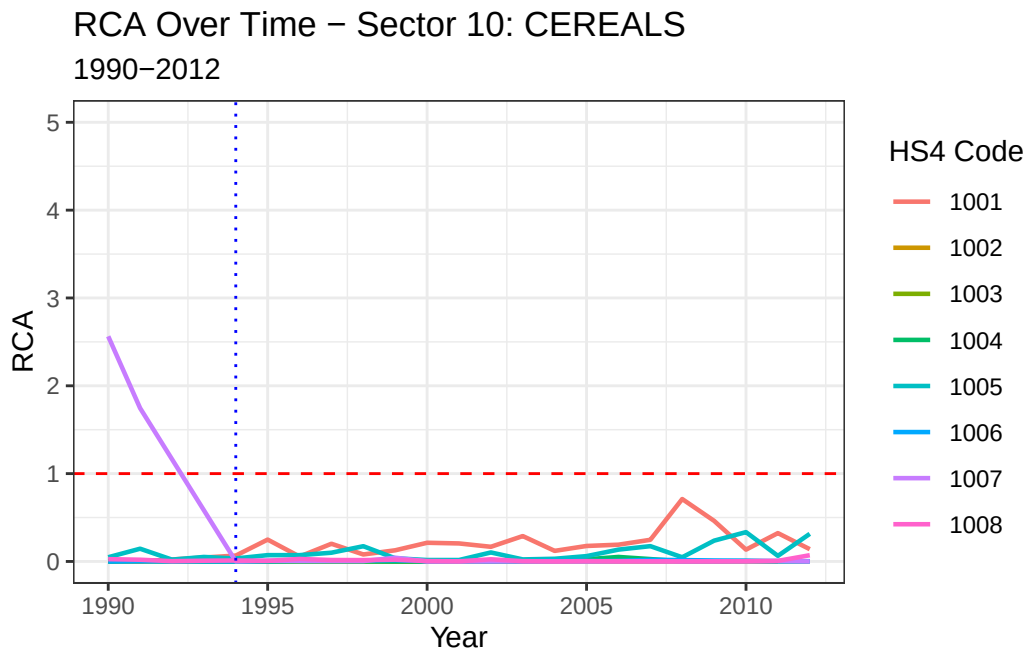
10 CEREALS

No 4 digit HS classification code is omitted.

```
rca_sector_10 <- rca_big_df |>
  filter(sector == "10", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_10, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 10: CEREALS",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



11 Milling products; malt; starch, etc.; gluten

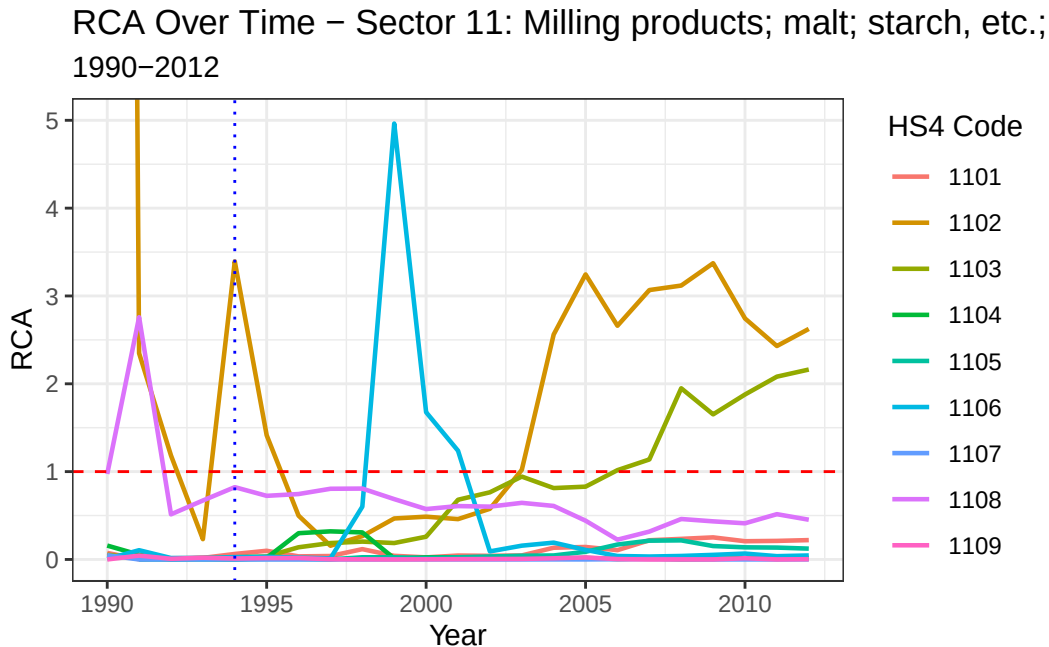
No 4 digit HS classification code is omitted.

No 4 digit HS classification code is omitted.

```
rca_sector_11 <- rca_big_df |>
  filter(sector == "11", nchar(cmdCode) == 4)
```

Plot sector 11

```
ggplot(rca_sector_11, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 11: Milling products; malt; starch, etc.; gluten",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



12 Oilseeds; kernels; industrial medicinal plants; feed

We omit the following HS Codes:

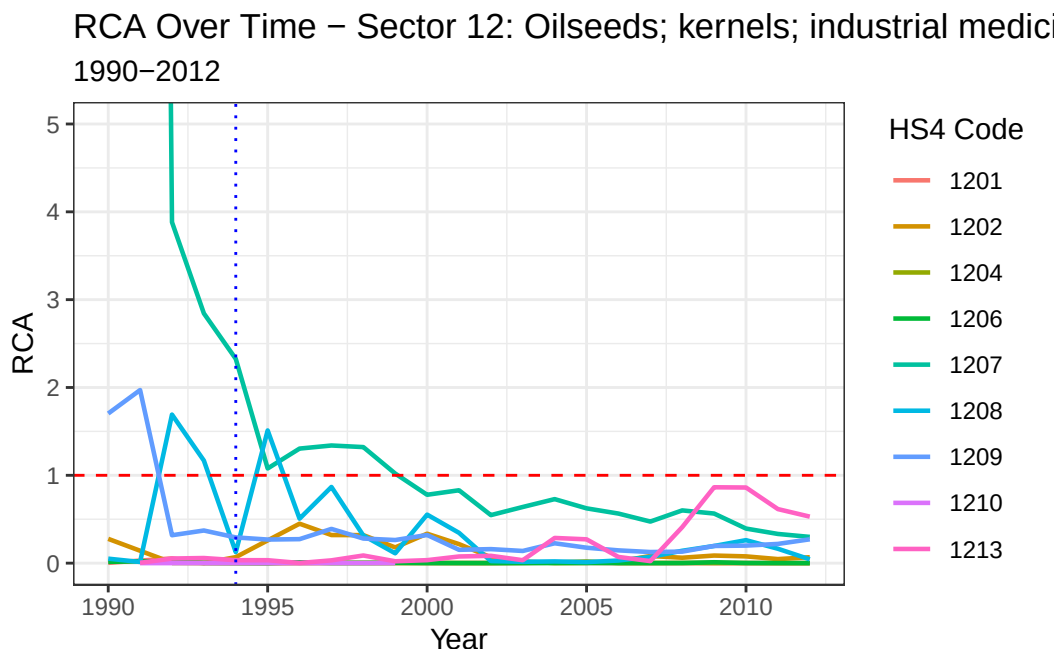
- 1205 Rape or colza seeds; whether or not broken
- 1211 Plants and parts of plants (including seeds and fruits), of a kind used primarily in perfumery, in pharmacy or for insecticidal, fungicidal or similar purposes, fresh, chilled, frozen or dried, whether or not cut, crushed or powdered
- 1212 Locust beans, seaweeds and other algae, sugar beet, sugar cane, fresh, chilled, frozen or dried, whether or not ground; fruit stones, kernels and other vegetable products (including unroasted chicory roots) used primarily for human consumption, n.e.c.

- 1214 Swedes, mangolds, fodder roots, hay, lucerne (alfalfa), clover, sainfoin, forage kale, lupines, vetches and similar forage products, whether or not in the form of pellets

```
rca_sector_12 <- rca_big_df |>
  filter(sector == "12", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_12, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 12: Oilseeds; kernels; industrial medicinal plants; f",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

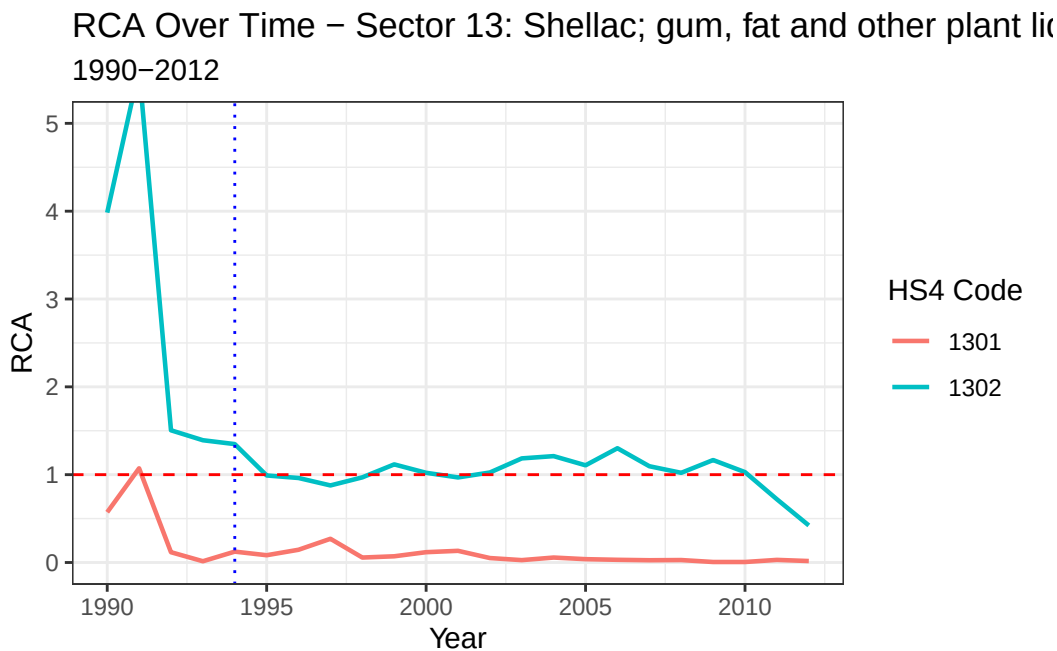


13 Shellac; gum, fat and other plant liquids, juice

```
rca_sector_13 <- rca_big_df |>
  filter(sector == "13", nchar(cmdCode) == 4)
```

Plot sector 13

```
ggplot(rca_sector_13, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 13: Shellac; gum, fat and other plant liquids, juice"
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



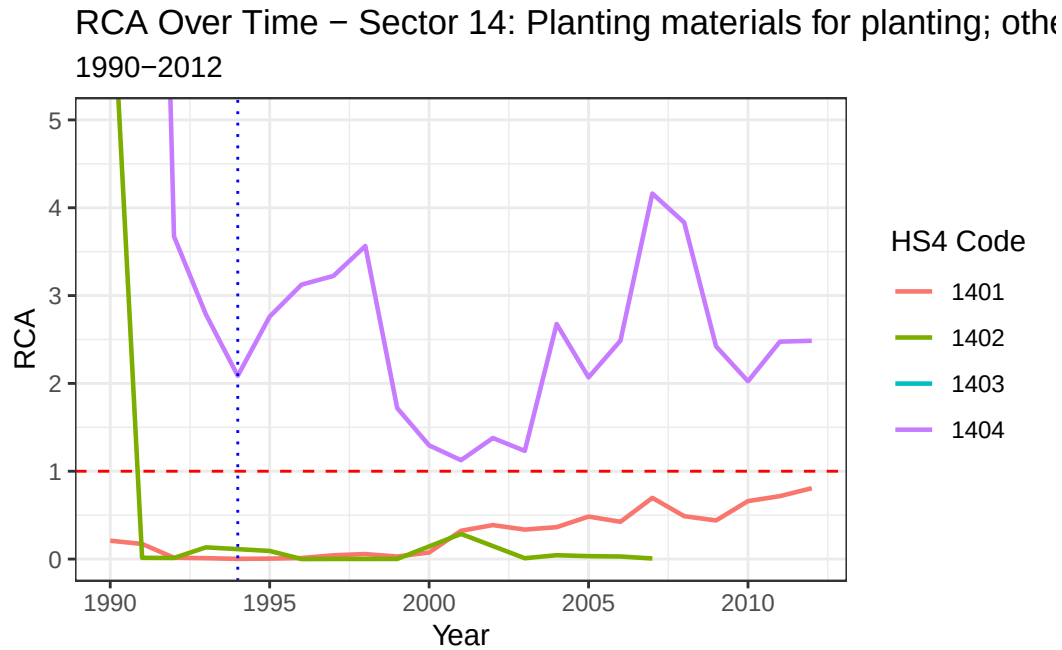
14 Planting materials for planting; other plant products

```
rca_sector_14 <- rca_big_df |>
  filter(sector == "14", nchar(cmdCode) == 4)
```

Plot sector 14

```
ggplot(rca_sector_14, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
```

```
geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
labs(title = "RCA Over Time - Sector 14: Planting materials for planting; other plant products",
      subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
theme_bw()
```



15 Animal and vegetable oils and fats; refined edible oils and fats

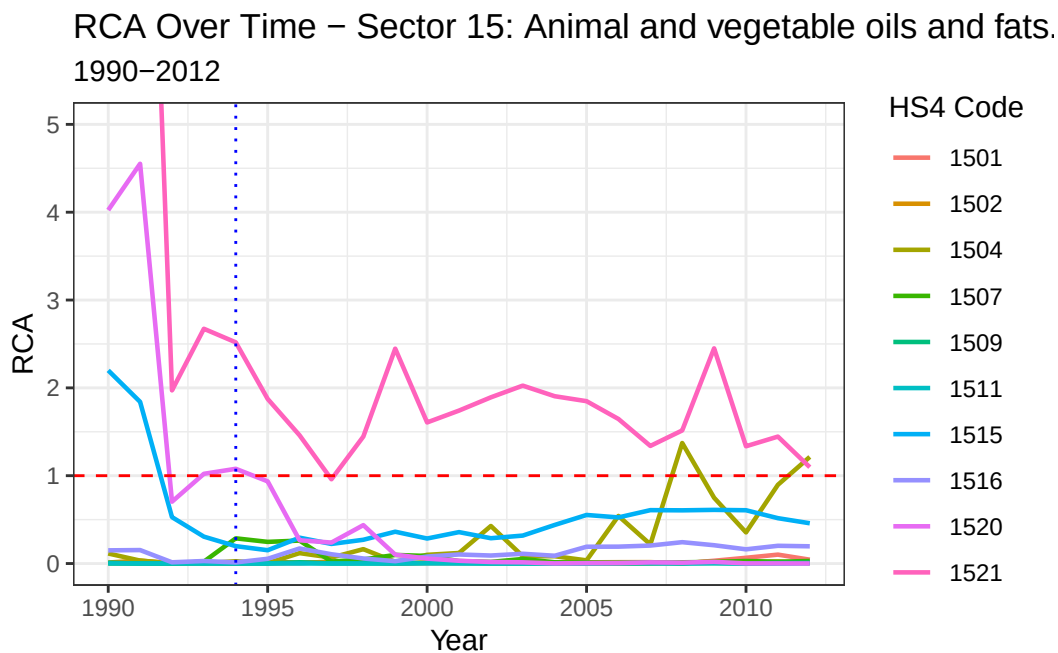
We Omit the following:

- 1503
- 1505
- 1506
- 1508
- 1510
- 1511 to 1514
- 1517 to 1519


```
rca_sector_15 <- rca_big_df |>
  filter(sector == "15", nchar(cmdCode) == 4)
```

Plot sector 15

```
ggplot(rca_sector_15, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 15: Animal and vegetable oils and fats...",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



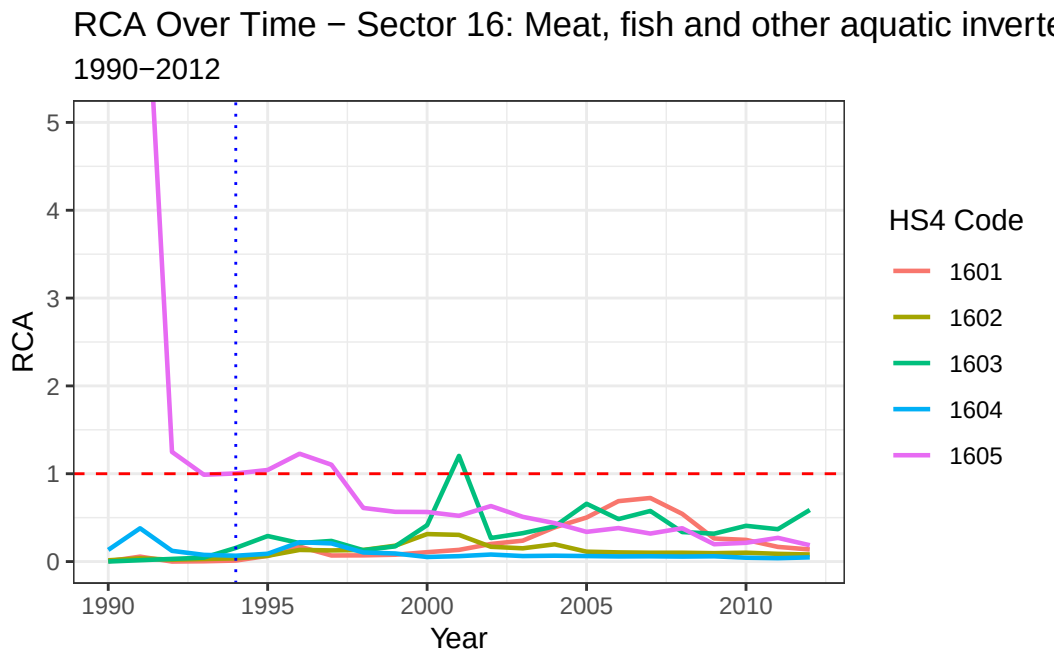
16 Meat, fish and other aquatic invertebrate products

RCA Sector 16

```
rca_sector_16 <- rca_big_df |>
  filter(sector == "16", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_16, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 16: Meat, fish and other aquatic invertebrate products",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



17 Sugar and confectionery

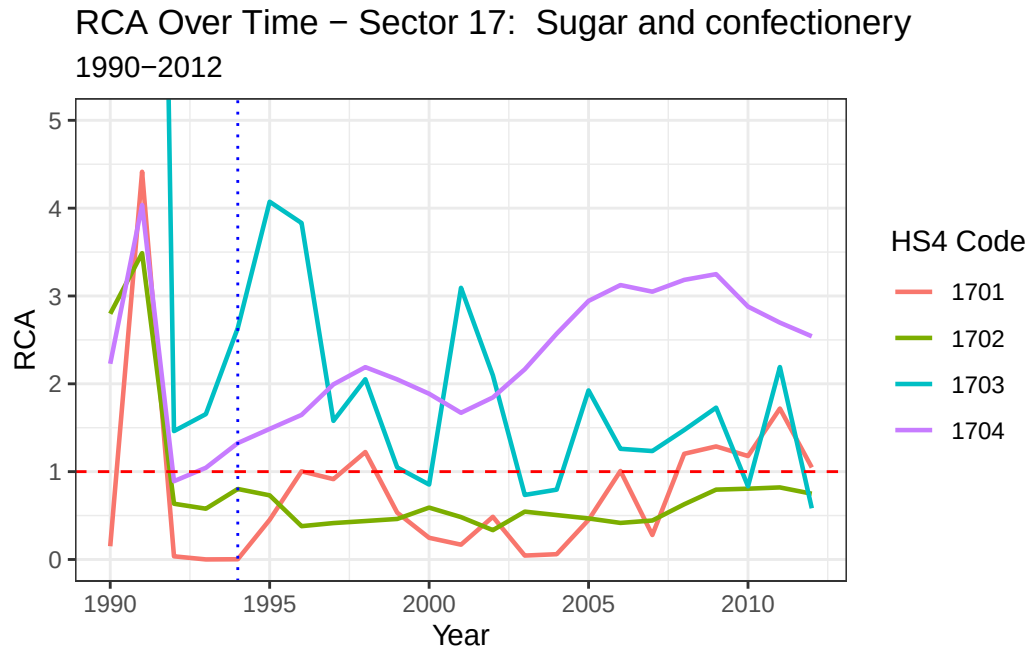
RCA Sector 17

```
rca_sector_17 <- rca_big_df |>
  filter(sector == "17", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_17, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
```

```
coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
labs(title = "RCA Over Time - Sector 17: Sugar and confectionery",
      subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
theme_bw()
```



18 Cocoa and cocoa products

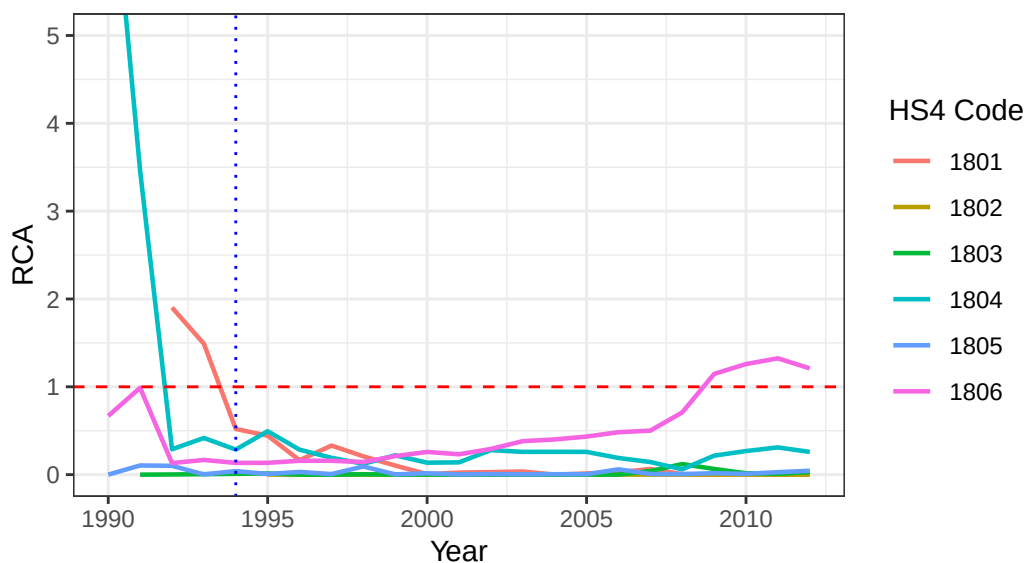
RCA Sector 18

```
rca_sector_18 <- rca_big_df |>
  filter(sector == "18", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_18, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 18: Cocoa and cocoa products",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

RCA Over Time – Sector 18: Cocoa and cocoa products 1990–2012



19 Cereal flour, starch, etc. or dairy products; cakes

RCA Sector 19

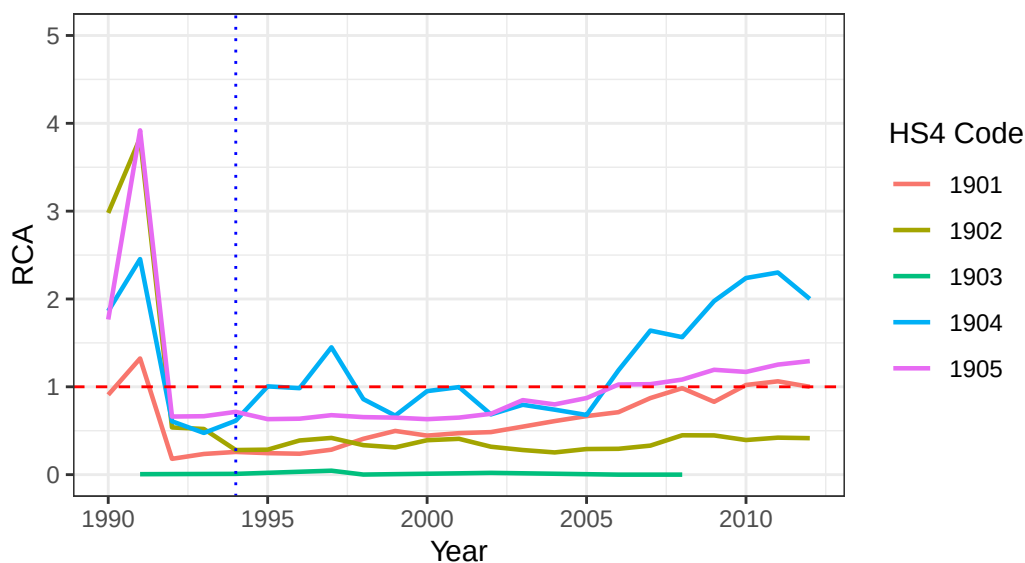
No HS Code is omitted

```
rca_sector_19 <- rca_big_df |>
  filter(sector == "19", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_19, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 19: Cereal flour, starch, etc. or dairy products; cakes",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

RCA Over Time – Sector 19: Cereal flour, starch, etc. or dairy pr 1990–2012



20 Products of vegetables, fruits or other parts of plants

RCA Sector 20

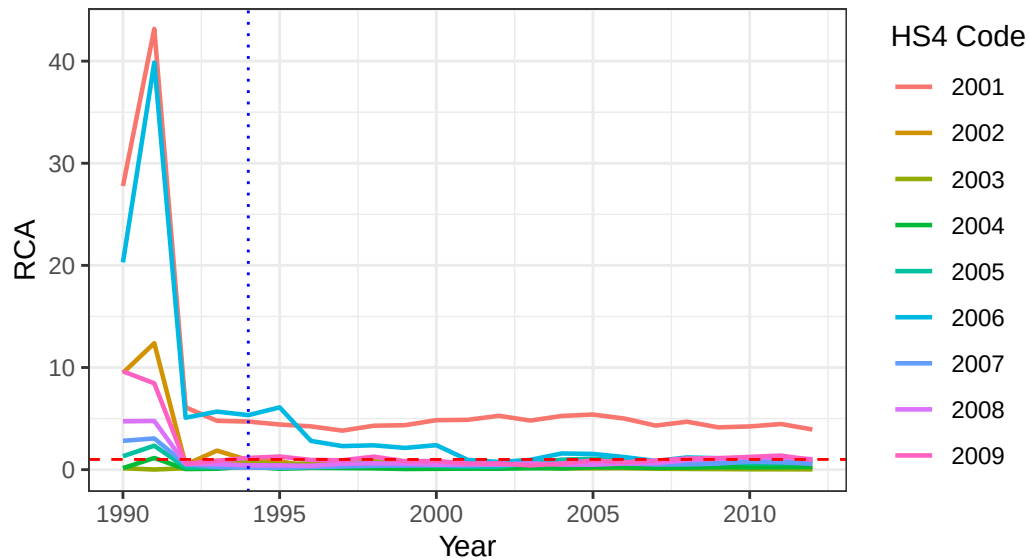
No HS Code is omitted

```
rca_sector_20 <- rca_big_df |>
  filter(sector == "20", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_20, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 43)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 20: Products of vegetables, fruits or other parts of p",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

RCA Over Time – Sector 20: Products of vegetables, fruits or o 1990–2012



21 Miscellaneous food

RCA Sector 21

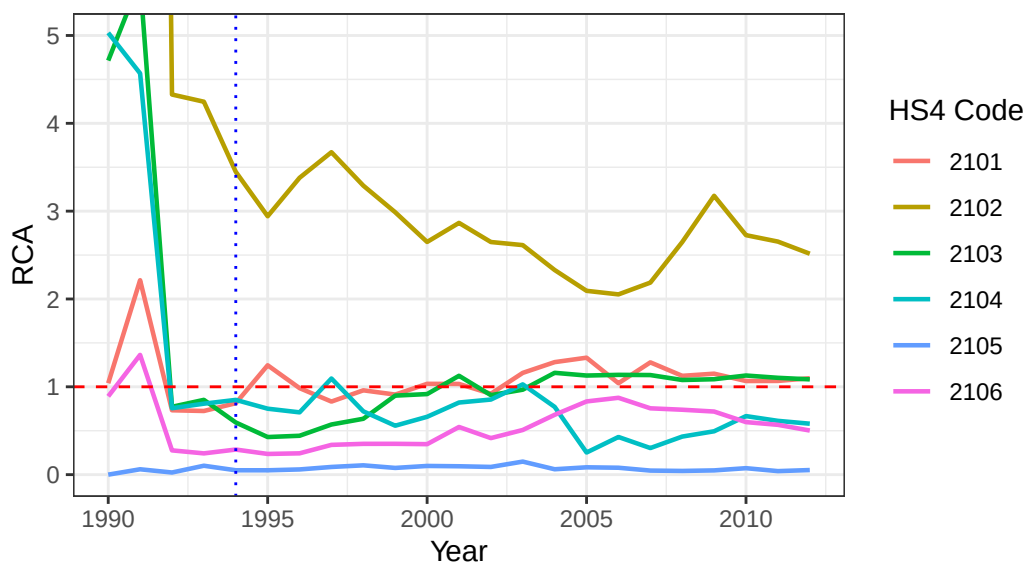
No HS Code is omitted

```
rca_sector_21 <- rca_big_df |>
  filter(sector == "21", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_21, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Miscellaneous food",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

RCA Over Time – Miscellaneous food 1990–2012



22 Beverages, wine and vinegar

RCA Sector 22

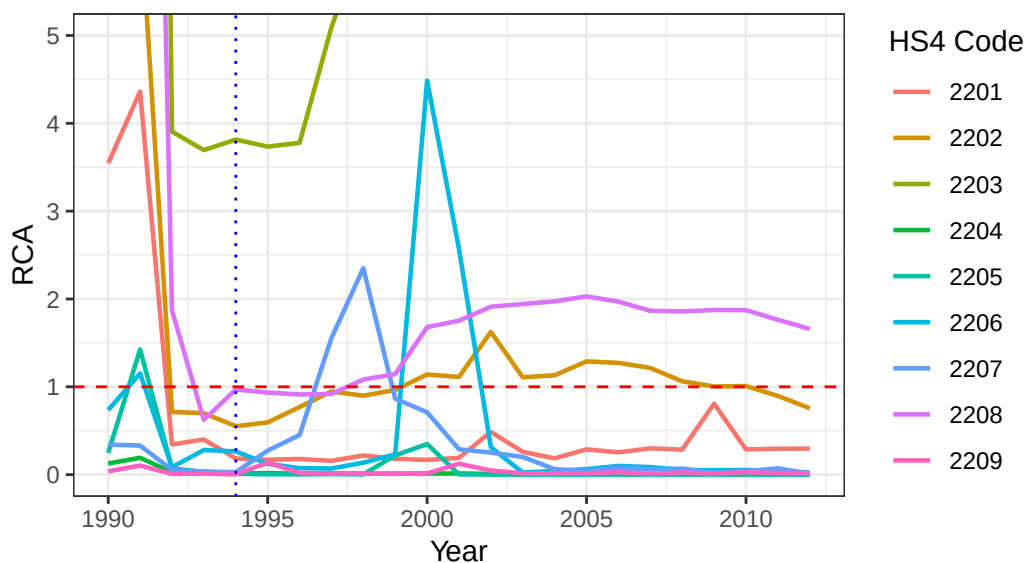
No HS Code is omitted

```
rca_sector_22 <- rca_big_df |>
  filter(sector == "22", nchar(cmdCode) == 4)
```

Plot sector 22

```
ggplot(rca_sector_22, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 22 Beverages, wine and vinegar",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

RCA Over Time – Sector 22 Beverages, wine and vinegar 1990–2012



23 Food industry residues and waste; formulated feed

RCA Sector 23

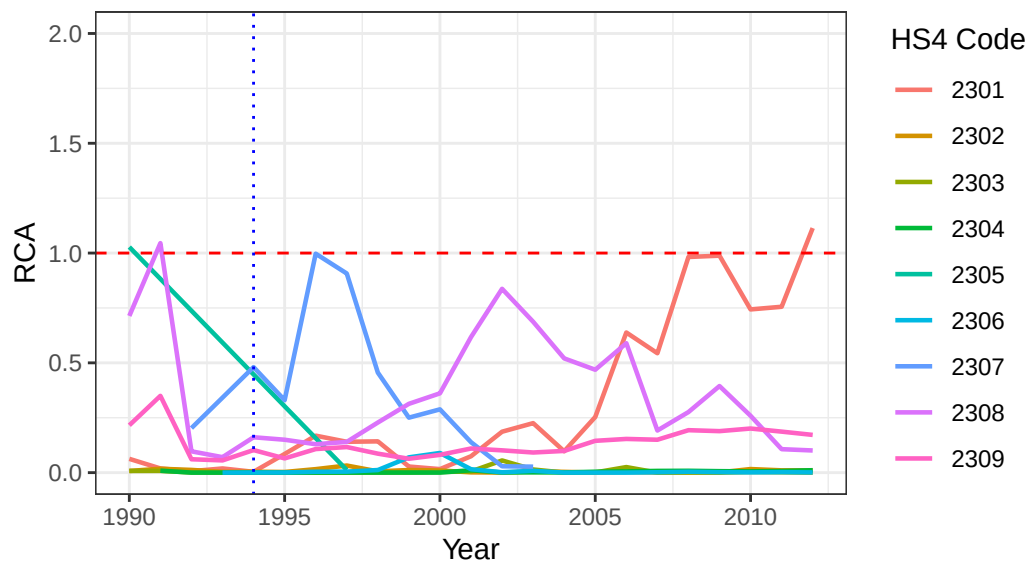
No HS Code is omitted

```
rca_sector_23 <- rca_big_df |>
  filter(sector == "23", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_23, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 2)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 23: Food industry residues and waste; formulated feed",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```


RCA Over Time – Sector 23: Food industry residues and waste 1990–2012



24 Tobacco, tobacco and tobacco substitute products

RCA Sector 24

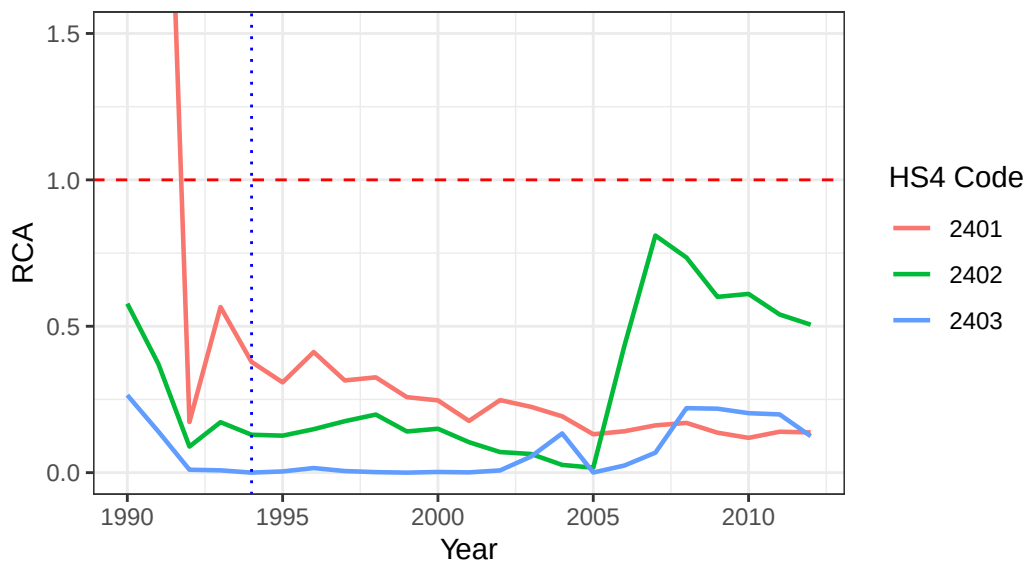
No HS Code is omitted.

```
rca_sector_24 <- rca_big_df |>
  filter(sector == "24", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_24, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 1.5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 24: Tobacco, tobacco and tobacco substitute products",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

RCA Over Time – Sector 24: Tobacco, tobacco and tobacco su 1990–2012



50 Silk

RCA Sector 50

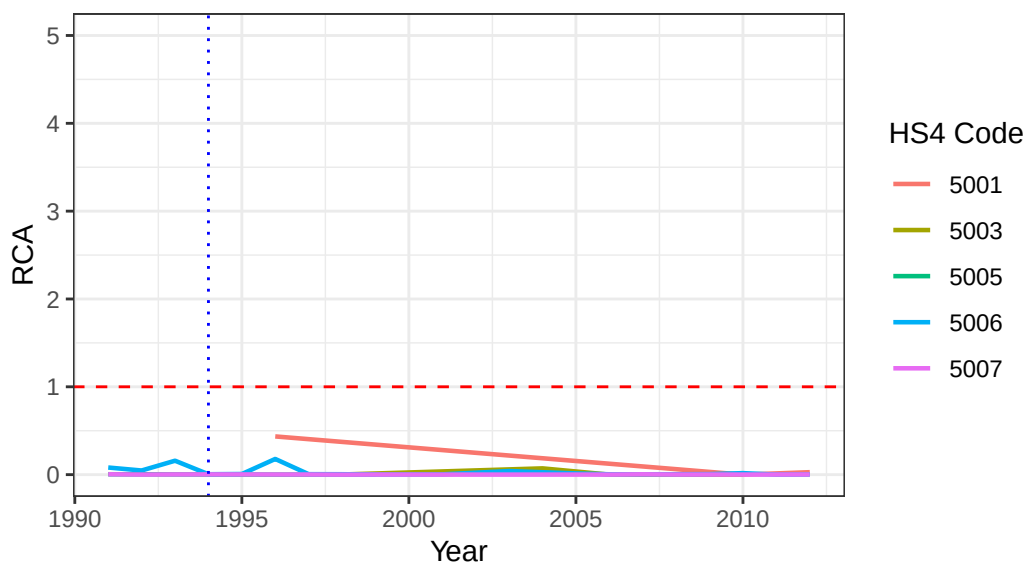
No HS Code is omitted

```
rca_sector_50 <- rca_big_df |>
  filter(sector == "50", nchar(cmdCode) == 4)
```

Plot sector 50

```
ggplot(rca_sector_50, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 50: Silk",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

RCA Over Time – Sector 50: Silk 1990–2012



51 Wool and other animal hair

RCA Sector 51

We omit the following HS Codes

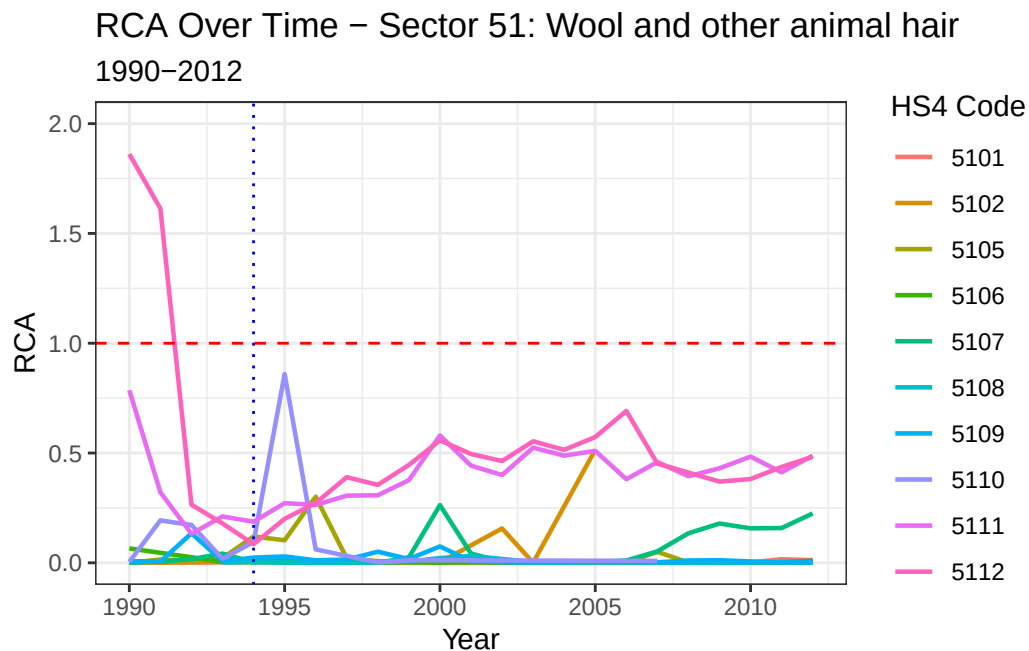
- 5103 Waste of wool or of fine or coarse animal hair, including yarn waste but excluding garnetted stock
- 5104 Wool, or fine or coarse animal hair; garnetted stock
- 5113 Woven fabrics of coarse animal hair or of horsehair

```
rca_sector_51 <- rca_big_df |>
  filter(sector == "51", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_51, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 2)) + # cap at 10, adjust as needed
```

```
labs(title = "RCA Over Time - Sector 51: Wool and other animal hair",
     subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
theme_bw()
```



52 Cotton

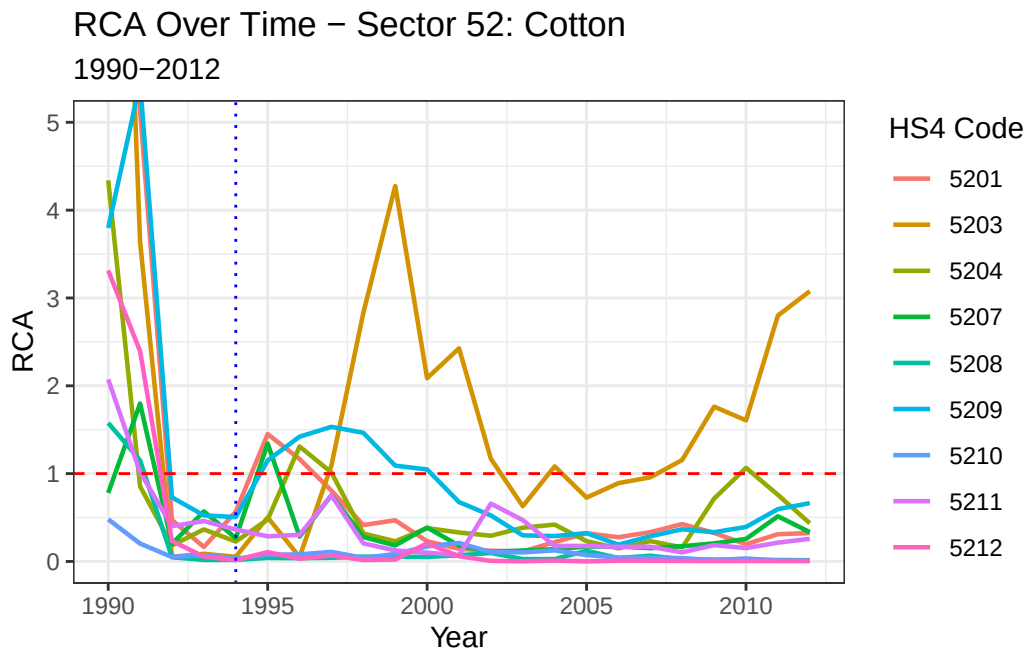
Omit

- 5205 Cotton yarn (other than sewing thread), containing 85% or more by weight of cotton, not put up for retail sale
- 5206 Cotton yarn (other than sewing thread), containing less than 85% by weight of cotton, not put up for retail sale

```
rca_sector_52 <- rca_big_df |>
  filter(sector == "52", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_52, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 52: Cotton",
        subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



53 Other plant fibers

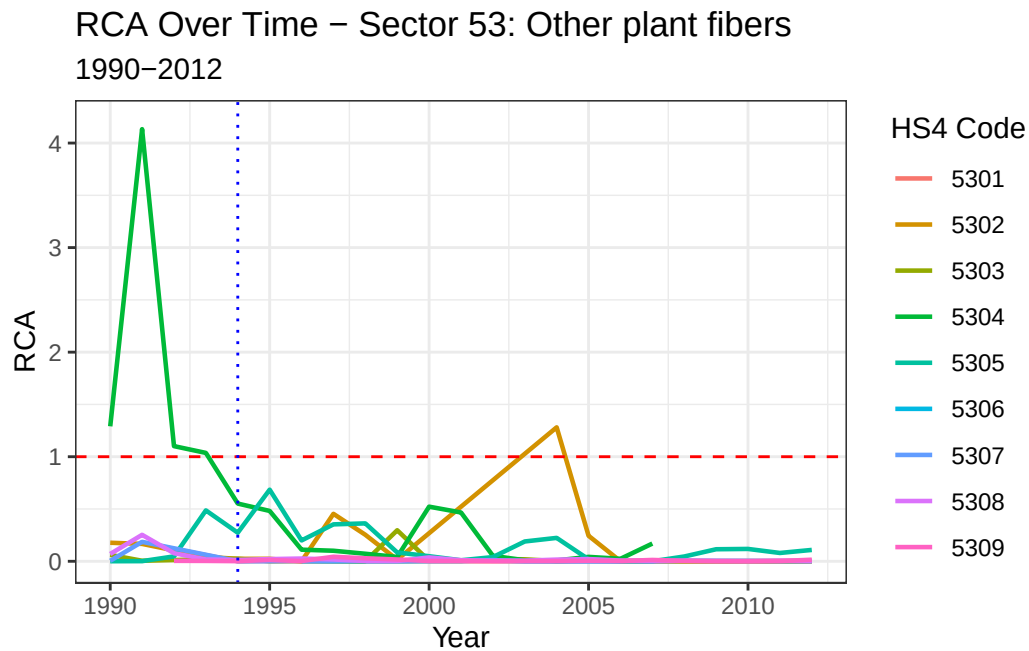
We omit the following:

- 5310 Woven fabrics of jute, other textile bast fibres of heading no. 5303
- 5311 Woven fabrics of other vegetable textile fibres; woven fabrics of paper yarn

```
rca_sector_53 <- rca_big_df |>
  filter(sector == "53", nchar(cmdCode) == 4)
```

Plot sector 53

```
ggplot(rca_sector_53, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 4.2)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 53: Other plant fibers",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



Load Variables for Regression

```
# load exchange rate dataset and tidy up
exch_rate <- read_csv("regression/variables/exch_rates.csv")

# (https://databank.worldbank.org/reports.aspx?source=2&series=PA.NUS.FCRF&country=MEX)

# load dataset for fdi inflow and tidy up

fdi_inflow <- read_csv("regression/variables/fdi_inflow.csv") # worldbank filtered out
```

```

fdi_inflow <- fdi_inflow |>
  select(Time, `Foreign direct investment, net inflows (BoP, current US$) [BX.KLT.DINV.CD.
  rename(period = "Time", fdiinflow = `Foreign direct investment, net inflows (BoP, current U
  na.omit(fdi_inflow)

# load datasets for agriculture fdi

agriculture_fdi_pt_1 <- read_csv("regression/variables/agriculture_fdi_pt_1.csv") ## https://

fdi_agri_mexico_2 <- read_csv("regression/variables/fdi_agri_mexico_2.csv")

agriculture_fdi_pt_1 <- agriculture_fdi_pt_1 |>
  select(period, agri_fdi)

agriculture_fdi_pt_2 <- fdi_agri_mexico_2 |>
  filter(anio >= 2001 & anio <= 2012) |>
  mutate(fn_millones_de_dolares = as.numeric(fn_millones_de_dolares)) |>
  group_by(anio) |>
  summarise(total_agri_fdi = sum(fn_millones_de_dolares, na.rm = TRUE), .groups = "drop") |>
  rename(period = anio)

agriculture_fdi_total <- agriculture_fdi_pt_1 |>
  mutate(agri_fdi = agri_fdi / 1e6) |>
  rename(total_agri_fdi = agri_fdi) |>
  bind_rows(agriculture_fdi_pt_2) |>
  arrange(period)

agriculture_fdi_total

```

```

# A tibble: 23 x 2
  period total_agri_fdi
  <dbl>         <dbl>
1  1990          61.1
2  1991          44.9
3  1992          39.3
4  1993          34.4
5  1994          10.3
6  1995           10
7  1996          31.7
8  1997           9.5

```

```
9    1998          27.6
10   1999          85.5
# i 13 more rows
```

Reg/EDA Df

```
## Create main Dataframe
rca_panel <- rca_big_df |>
  left_join(fdi_inflow, by = "period") |>
  left_join(exch_rate, by = "period") |>
  left_join(agriculture_fdi_total, by = "period") |>
  filter(nchar(cmdCode) == 4)

## store all cmdCodes as a vector

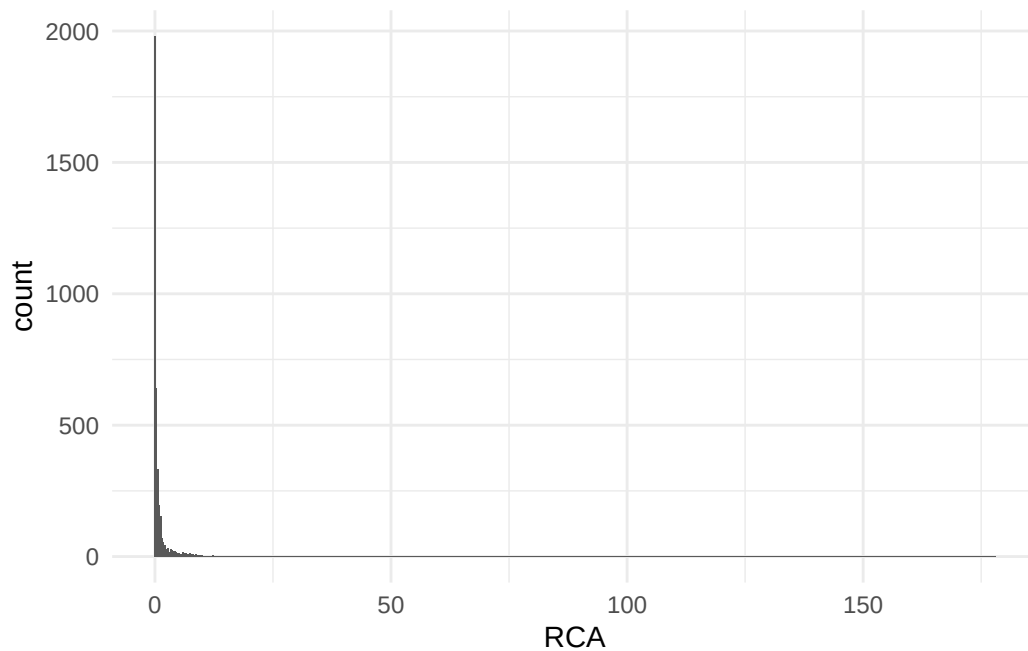
hs4_codes <- rca_panel |>
  pull(cmdCode)
```

EDA

Now that we have our main dataframe `rca_panel`, we can conduct an EDA to see whether or not we need to do any transformations before we model.

1) Distribution

```
rca_panel |>
  select(RCA) |>
  ggplot(aes(x = RCA)) +
  geom_histogram(binwidth = 0.3) +
  theme_minimal()
```

We observe that our RCA is heavily right skewed. Hence, we *might* have to do a $\log(\text{RCA})$ transformation due to the nature of the distribution.

2) Time

Show the time frame (1992-2000)

```
rca_panel|>
  distinct(period) |>
  arrange(period)
```

```
# A tibble: 23 x 1
  period
  <dbl>
1  1990
2  1991
3  1992
4  1993
5  1994
6  1995
7  1996
8  1997
9  1998
```

```
10 1999
# i 13 more rows
```

3) Num of products

```
rca_panel |>
  distinct(cmdCode) |> #gives us distinct values for cmdCode
  nrow() #gives us the number of rows
```

```
[1] 194
```

The number of HSCodes that we are analyzing is 191.

4) Balanced

Does every product appear in every year?

```
rca_panel |>
  count(cmdCode) |>
  summary()
```

cmdCode	n
Length:194	Min. : 2.00
Class :character	1st Qu.:19.25
Mode :character	Median :23.00
	Mean :20.25
	3rd Qu.:23.00
	Max. :23.00

5) Correlations

We must check for correlations between our explanatory variables

```
rca_panel |>
  distinct(period, post_nafta, exch_rate, fdiinflow, total_agri_fdi) |>
  select(post_nafta, exch_rate, total_agri_fdi )|>
  cor()
```

	post_nafta	exch_rate	total_agri_fdi
post_nafta	1.00000000	0.77167205	-0.02806659
exch_rate	0.77167205	1.00000000	0.09090548
total_agri_fdi	-0.02806659	0.09090548	1.00000000

6) Transform the variables

```
rca_panel <- rca_panel|>
  mutate(
    log_rca = log(RCA), # if RCA is right-skewed
    log_exch = log(exch_rate),
    asinh_fdi = asinh(total_agri_fdi) # sinh = (e^x - e^-x)/2
  )
```

7) Num of Prods RCA > 1

```
rca_greq1 <- rca_panel |>
  filter(RCA >= 1)

## what are the products with RCA >= 1? regardless of time

rca_greq1 |>
  select(cmdCode) |>
  unique()
```

```
# A tibble: 95 x 1
  cmdCode
  <chr>
1 0102
2 0101
3 0202
4 0205
5 0207
6 0209
7 0201
8 0502
9 0503
10 0508
# i 85 more rows
```

8) Products which did not start off with comparative advantage before nafta, and ended up with one after?

```
#products that did not have a CA before NAFTA
no_ca_before <- rca_panel |>
  filter(post_nafta == 0) |> ## before nafta
  group_by(cmdCode) |> #group by commodity Code
  summarise(had_ca = any(has_CA)) |> # if any values in has_CA is true, i.e does have a CA
  filter(had_ca == FALSE) |>
  pull(cmdCode)

no_ca_before
```

```
[1] "0103" "0104" "0105" "0201" "0203" "0206" "0208" "0209" "0210" "0401"
[11] "0402" "0403" "0404" "0405" "0406" "0407" "0408" "0410" "0501" "0504"
[21] "0505" "0506" "0507" "0701" "0801" "0808" "0809" "0902" "0903" "0906"
[31] "0907" "0908" "0909" "0910" "1001" "1003" "1004" "1005" "1006" "1008"
[41] "1101" "1103" "1104" "1105" "1106" "1107" "1109" "1201" "1202" "1204"
[51] "1206" "1210" "1213" "1401" "1501" "1502" "1504" "1507" "1509" "1511"
[61] "1516" "1601" "1602" "1603" "1604" "1803" "1805" "1806" "1903" "2003"
[71] "2105" "2204" "2207" "2209" "2301" "2302" "2303" "2304" "2306" "2307"
[81] "2309" "2402" "2403" "5005" "5006" "5007" "5101" "5102" "5105" "5106"
[91] "5107" "5108" "5109" "5110" "5111" "5210" "5301" "5302" "5303" "5305"
[101] "5307" "5308" "5309"
```

```
# of those products which ones developed a comparative advantage after 1994?
gained_ca_after <- rca_panel |>
  filter(post_nafta == 1, cmdCode %in% no_ca_before) |>
  group_by(cmdCode) |>
  summarise(gained_ca = any(has_CA)) |>
  filter(gained_ca == TRUE) |>
  pull(cmdCode)

gained_ca_after
```

```
[1] "0201" "0209" "1103" "1106" "1504" "1603" "1806" "2207" "2301" "5302"
```

9) Descriptive overview

```
rca_panel |>
  group_by(post_nafta) |>
  summarise(
    mean_rca = mean(RCA, na.rm = TRUE),
    median_rca = median(RCA, na.rm = TRUE),
    sd_rca = sd(RCA, na.rm = TRUE),
    min_rca = min(RCA, na.rm = TRUE),
    max_rca = max(RCA, na.rm = TRUE),
    pct_with_RCA = mean(has_CA)*100
  )
```

```
# A tibble: 2 x 7
  post_nafta mean_rca median_rca sd_rca      min_rca max_rca pct_with_RCA
    <dbl>      <dbl>      <dbl> <dbl>      <dbl>    <dbl>      <dbl>
1         0     4.61       0.203  15.3  0.000000264    178.        32.9
2         1     0.836       0.136   2.19  0.0000000326    37.2        18.5
```

10) Prevalence in RCA by sector

```
rca_panel |>
  group_by(sector, post_nafta) |>
  summarise(
    n_products= n_distinct(cmdCode), #counts num of distinct values
    pct_with_CA = mean(has_CA) * 100,
    mean_rca = mean(RCA, na.rm = TRUE)
  )
```

```
# A tibble: 52 x 5
# Groups:   sector [26]
  sector post_nafta n_products pct_with_CA mean_rca
  <chr>      <dbl>      <int>      <dbl>      <dbl>
1 01         0         5        29.4       7.53
2 01         1         5        23.8       0.935
3 02         0         9        20.6       0.743
4 02         1        10         6.67       0.326
5 04         0         9         0        0.0155
6 04         1         9         0        0.0454
```

```

7 05      0      9      28.1    3.33
8 05      1      9       7.26    0.458
9 07      0     10      85      23.7
10 07     1     10     73.2     3.89
# i 42 more rows

```

11) Year by Year

```

rca_panel |>
  group_by(period) |>
  summarise(
    n_with_CA = sum(has_CA),
    total_products = n_distinct(cmdCode),
    pct_with_CA = mean(has_CA)*100
  )

```

```

# A tibble: 23 x 4
  period n_with_CA total_products pct_with_CA
  <dbl>   <int>         <int>     <dbl>
1  1990      72          166      43.4
2  1991      79          168      47.0
3  1992      31          158      19.6
4  1993      32          159      20.1
5  1994      31          167      18.6
6  1995      34          167      20.4
7  1996      29          174      16.7
8  1997      29          173      16.8
9  1998      29          173      16.8
10 1999      28          176      15.9
# i 13 more rows

```

12) Volatility of RCA

```

rca_panel |>
  arrange(cmdCode, period) |>
  group_by(cmdCode) |>
  summarise(
    switches = sum(has_CA != lag(has_CA), na.rm = TRUE)
  ) |>

```

```
filter(switches > 1) |>
arrange(desc(switches))
```

```
# A tibble: 49 x 2
  cmdCode switches
  <chr>      <int>
1 1701         9
2 0802         8
3 1703         7
4 2009         7
5 0901         6
6 1904         6
7 2101         6
8 5203         6
9 1302         5
10 2006         5
# i 39 more rows
```

13 Regressor variation

```
rca_panel |>
  group_by(nafta_phase) |>
  summarise(
    mean_exch_rate = mean(exch_rate, na.rm = TRUE),
    mean_fdi = mean(fdiinflow, na.rm = TRUE),
    mean_agri_fdi = mean(total_agri_fdi, na.rm = TRUE),
    n_obs = n()
  )
```

```
# A tibble: 3 x 5
  nafta_phase mean_exch_rate mean_fdi mean_agri_fdi n_obs
  <chr>      <dbl>      <dbl>      <dbl> <int>
1 post          7.66 12497714312.      38.7  1195
2 pre           3.01  4038717358.      45.1   651
3 <NA>          11.4      NaN      44.1  2083
```

14) RCA Trajectories

```
get_rca_trajectory <- function(tempcmdcde) {  
  rca_panel |>  
    filter(cmdCode == tempcmdcde) |>  
    select(period, cmdCode, RCA, has_CA, exch_rate, fdiinflow) |>  
    arrange(period) |>  
    mutate(rca_change = RCA - lag(RCA)) # how much RCA shifted each year  
}  
  
map(unique(rca_panel$cmdCode), get_rca_trajectory) |>  
  list_rbind()
```

```
# A tibble: 3,929 x 7  
  period cmdCode   RCA has_CA exch_rate fdiinflow rca_change  
  <dbl> <chr>   <dbl> <lgl>   <dbl>     <dbl>     <dbl>  
1  1990 0101   0.804 FALSE    2.81 2634000000    NA  
2  1991 0101   1.22  TRUE    3.02 4762000000   0.416  
3  1992 0101   0.268 FALSE    3.09 4393000000  -0.952  
4  1993 0101   0.204 FALSE    3.12 4389000000  -0.0635  
5  1994 0101   0.0470 FALSE    3.38 10972500000 -0.157  
6  1995 0101   0.236 FALSE    6.42  9526290000   0.189  
7  1996 0101   0.125 FALSE    7.60  9185600000  -0.111  
8  1997 0101   0.0370 FALSE    7.92 12829800000  -0.0875  
9  1998 0101   0.0747 FALSE    9.14 12756764558   0.0376  
10 1999 0101   0.0488 FALSE    9.56 13941043232  -0.0258  
# i 3,919 more rows
```

15) How long did it take for products to gain RCA after NAFTA?

```
rca_panel |>  
  filter(post_nafta == 1, has_CA == TRUE) |>  
  group_by(cmdCode) |>  
  summarise(first_ca_year = min(period)) |>  
  mutate(years_after_nafta = first_ca_year - 1994) |>  
  arrange(years_after_nafta) |>  
  arrange(desc(years_after_nafta))
```

```
# A tibble: 63 x 3
```


	cmdCode	first_ca_year	years_after_nafta
	<chr>	<dbl>	<dbl>
1	0201	2012	18
2	2301	2012	18
3	1901	2010	16
4	1806	2009	15
5	2008	2009	15
6	1504	2008	14
7	0705	2007	13
8	1103	2006	12
9	1905	2006	12
10	2005	2005	11

i 53 more rows

16) Consistency in RCAS

```
rca_panel |>
  filter(post_nafta == 1) |>
  group_by(cmdCode) |>
  summarise(
    years_with_CA = sum(has_CA),
    total_years = n(),
    consistency = years_with_CA/total_years
  ) |>
  mutate(ca_type = case_when(
    consistency == 1 ~ "Persistent CA",
    consistency >= 0.5 ~ "Majority CA",
    consistency > 0 & consistency < 0.5 ~ "Sporadic CA",
    consistency == 0 ~ "No CA"
  )) |>
  count(ca_type)
```

```
# A tibble: 4 x 2
  ca_type      n
  <chr>    <int>
1 Majority CA    16
2 No CA        131
3 Persistent CA    16
4 Sporadic CA     31
```

17) Avg RCA By sector

```
rca_panel |>
  group_by(sector, post_nafta) |>
  summarise(
    mean_rca = mean(RCA),
    median_rca = median(RCA),
    n_products = n_distinct(cmdCode)
  ) |>
  pivot_wider(
    names_from = post_nafta,
    values_from = c(mean_rca, median_rca, n_products),
    names_prefix = "post_"
  )
```

A tibble: 26 x 7

Groups: sector [26]

	sector	mean_rca_post_0	mean_rca_post_1	median_rca_post_0	median_rca_post_1
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	01	7.53	0.935	0.0203	0.0116
2	02	0.743	0.326	0.0522	0.0551
3	04	0.0155	0.0454	0.00275	0.0148
4	05	3.33	0.458	0.511	0.112
5	07	23.7	3.89	7.13	3.28
6	08	11.0	1.92	1.86	0.850
7	09	3.49	0.269	0.156	0.0415
8	10	0.212	0.0525	0.00620	0.00576
9	11	1.98	0.474	0.0201	0.107
10	12	3.26	0.212	0.0516	0.0462

i 16 more rows

i 2 more variables: n_products_post_0 <int>, n_products_post_1 <int>

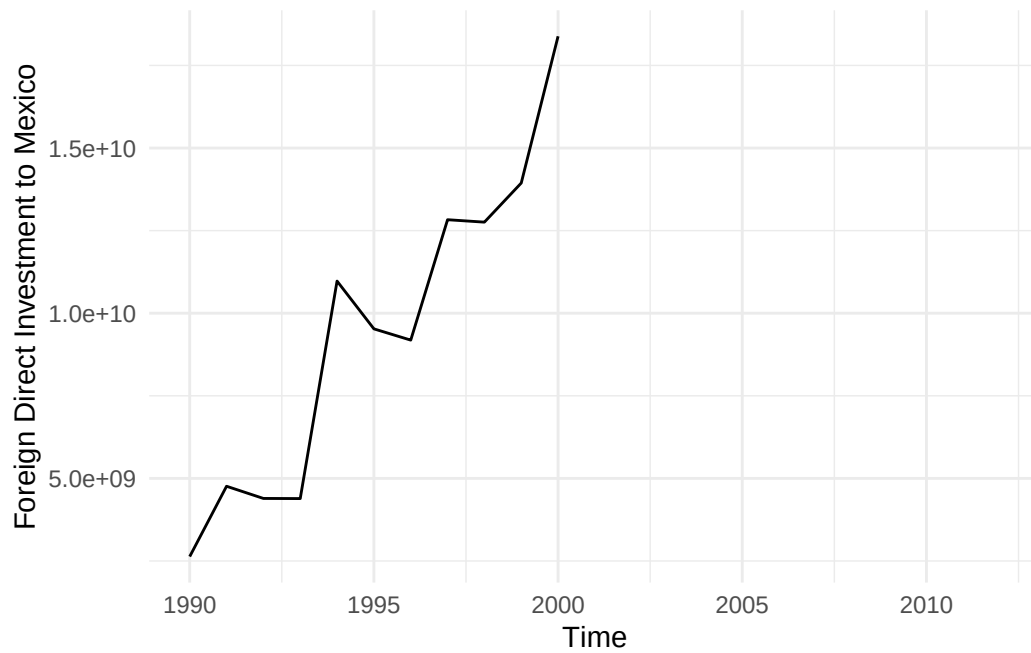
18) Largest RCA Gains?

```
rca_panel |>
  group_by(sector, post_nafta) |>
  summarise(mean_rca = mean(RCA)) |>
  pivot_wider(names_from = post_nafta, values_from = mean_rca, names_prefix = "period_") |>
  mutate(rca_gain = period_1 - period_0) |>
  arrange(desc(rca_gain))
```

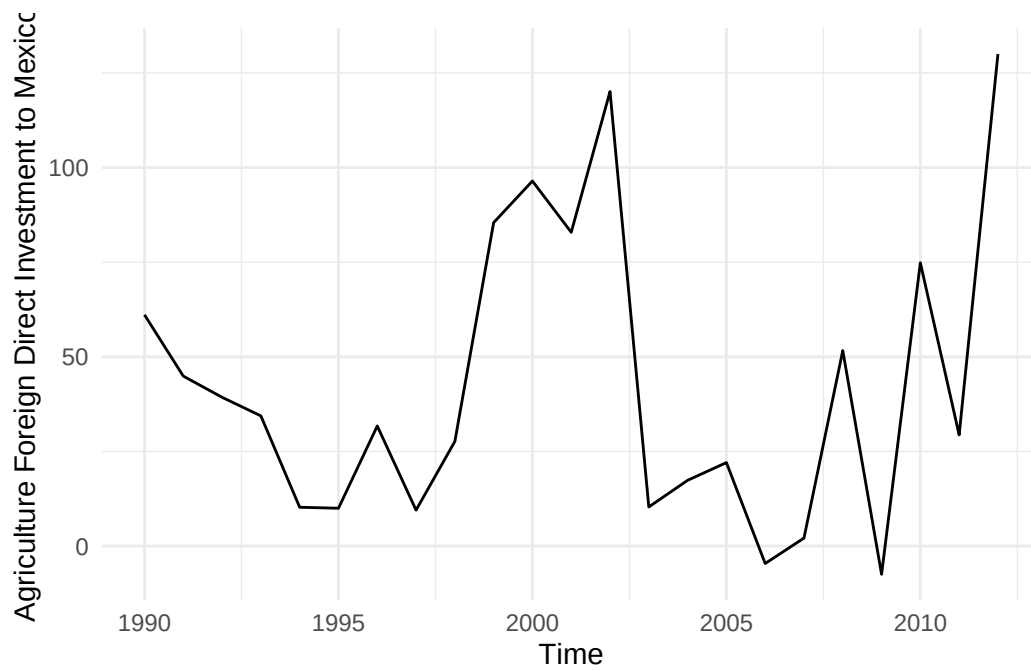
```
# A tibble: 26 x 4
# Groups:   sector [26]
  sector period_0 period_1 rca_gain
  <chr>      <dbl>    <dbl>    <dbl>
1 04         0.0155  0.0454   0.0299
2 23         0.161   0.147  -0.0135
3 50         0.0425  0.0190  -0.0236
4 51         0.175   0.125  -0.0507
5 10         0.212   0.0525  -0.159
6 53         0.407   0.0590  -0.348
7 02         0.743   0.326  -0.417
8 24         0.646   0.195  -0.451
9 19         1.35    0.702  -0.647
10 16         0.966   0.284  -0.683
# i 16 more rows
```

19) FDI over time

```
ggplot(data = rca_panel, aes(x = period, y = fdiinflow)) +
  geom_line() +
  theme_minimal() +
  xlab("Time") +
  ylab("Foreign Direct Investment to Mexico")
```



```
ggplot(data = rca_panel, aes(x = period, y = total_agri_fdi)) +
  geom_line() +
  theme_minimal() +
  xlab("Time") +
  ylab("Agriculture Foreign Direct Investment to Mexico")
```



MODELS

Model 1 Fixed Effects Regresission

```
model <- feols(RCA ~ post_nafta + log(exch_rate) + log(fdiinflow) | cmdCode,  
              data = rca_panel,  
              cluster = ~cmdCode)  
  
summary(model)
```

```
OLS estimation, Dep. Var.: RCA  
Observations: 1,846  
Fixed-effects: cmdCode: 194  
Standard-errors: Clustered (cmdCode)  
  
              Estimate Std. Error  t value  Pr(>|t|)  
post_nafta    -0.895031   0.425458  -2.10369 3.6701e-02 *  
log(exch_rate) 0.251852   0.174049   1.44702 1.4951e-01  
log(fdiinflow) -2.639237   0.661732  -3.98838 9.4364e-05 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
RMSE: 7.33774      Adj. R2: 0.328015  
                Within R2: 0.056776
```

Model 4 USE THIS

```
model4 <- feols(log(RCA) ~ post_nafta + log(exch_rate) + asinh(total_agri_fdi) | cmdCode,  
               data = rca_panel,  
               cluster = ~cmdCode)  
  
summary(model4)
```

```
OLS estimation, Dep. Var.: log(RCA)  
Observations: 3,929  
Fixed-effects: cmdCode: 194  
Standard-errors: Clustered (cmdCode)  
  
              Estimate Std. Error  t value  Pr(>|t|)  
post_nafta    -0.679558   0.152166  -4.465887 1.3554e-05 ***  
log(exch_rate) 0.084526   0.143496   0.589044 5.5652e-01
```

```

asinh(total_agri_fdi)  0.017943    0.010017  1.791247 7.4821e-02 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 1.50711      Adj. R2: 0.762785
                Within R2: 0.021981

```

Interpretation

The interpretation of model 4 will be as follows.

Log(Exchange Rate)

A 1 percent increase in the exchange rate is associated with a 0.43 percent increase in RCA (at the 1 percent level)

nafta__phaseearly

Log(RCA) was 0.41 (units) lower than in the pre-NAFTA baseline, holding exchange rate and agriculture_FDI constant (significant at the 5 percent level)

**** asinh(agri_fdi)****

A rise in agriculture FDI is associated with a decrease in RCA (counterintuitive).

A 1 percent increase in FDI in Mexico's Agriculture Sector is linked with a -0.142 percent decrease in RCA, holding all else constant.

- FDI may have been used to increase domestic production rather than building export competitiveness in the global market.
- FDI may flow into sectors that lose comparative advantage.
- Aggregate agriculture FDI may have been flowing disproportionately into subsectors (i.e fertilizer, etc..) so FDI is not actually reaching the products driving the RCA variation.

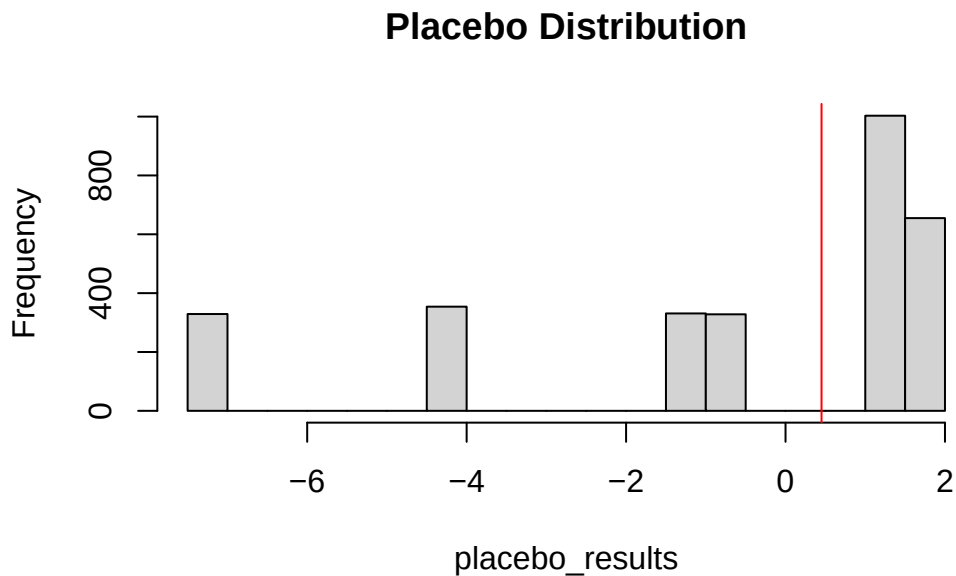
Monte Carlo Simulation Pseudo Code

```

set.seed(123)
placebo_results <- replicate(3000, {
  fake_year <- sample(1992:2000, 1)
  rca_panel$fake_nafta <- if_else(rca_panel$period >= fake_year, 1, 0)
  coef(feols(RCA ~ fake_nafta + log(exch_rate) + log(fdiinflow) | cmdCode,
             data = rca_panel))["fake_nafta"]
})

```

```
# Plot distribution of placebo coefficients
hist(placebo_results, main = "Placebo Distribution")
abline(v = 0.451646, col = "red")
```



TABLES

```
# Main Model Table (Claude AI)

model4_tidy <- broom::tidy(model4, conf.int = TRUE) |>
  mutate(
    term = case_when(
      term == "nafta_phaseearly" ~ "NAFTA Phase: Early",
      term == "nafta_phasepost" ~ "NAFTA Phase: Post",
      term == "log(exch_rate)" ~ "Log Exchange Rate",
      term == "asinh(agri_fdi)" ~ "IHS Agricultural FDI",
      TRUE ~ term
    ),
    stars = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.1 ~ "*",
      TRUE ~ ""
    ),
  ),
```

```

    estimate_stars = paste0(round(estimate, 3), stars),
    std.error = paste0("(", round(std.error, 3), ")")
  ) |>
  select(term, estimate_stars, std.error, p.value)

# Build gt table
model4_tidy |>
  gt() |>
  tab_header(
    title = "Fixed Effects Estimation of NAFTA's Impact on RCA",
    subtitle = "Dependent variable: Log(RCA)"
  ) |>
  cols_label(
    term = "Variable",
    estimate_stars = "Coefficient",
    std.error = "Std. Error",
    p.value = "p-value"
  ) |>
  fmt_number(
    columns = p.value,
    decimals = 3
  ) |>
  tab_spanner(
    label = "Log(RCA)",
    columns = c(estimate_stars, std.error, p.value)
  ) |>
  tab_footnote(
    footnote = "*** p < 0.01, ** p < 0.05, * p < 0.1" #shows statistical significance
  ) |>
  tab_footnote(
    footnote = "Clustered standard errors at the cmdCode level. Reference category: Pre-NAFTA"
  ) |>
  tab_source_note(
    source_note = "Source: UN Comtrade, Secretaría de Economía, Banco de México."
  ) |>
  tab_options(
    heading.title.font.size = 14,
    heading.subtitle.font.size = 12,
    column_labels.font.weight = "bold",
    table.width = pct(100)
  )

```


Fixed Effects Estimation of NAFTA's Impact on RCA

Dependent variable: Log(RCA)] Fixed Effects Estimation of NAFTA's Impact on RCA

	Variable	Log(RCA)	
		Coefficient	Std. Error
Dependent variable: Log(RCA)	post_nafta	-0.68***	(0.152)
	Log Exchange Rate	0.085	(0.143)
	asinh(total_agri_fdi)	0.018*	(0.01)

*** p < 0.01, ** p < 0.05, * p < 0.1

Clustered standard errors at the cmdCode level. Reference category: Pre-NAFTA.

Source: UN Comtrade, Secretaría de Economía, Banco de México.

Transition table

#Generated by Claude

```
transition_table <- rca_panel |>
  filter(post_nafta == 0) |>
  group_by(cmdCode) |>
  summarise(had_ca = any(has_CA)) |>
  inner_join(
    rca_panel |>
      filter(post_nafta == 1) |>
      group_by(cmdCode) |>
      summarise(gained_ca = any(has_CA)),
    by = "cmdCode"
  ) |>
  mutate(transition = case_when(
    !had_ca & gained_ca ~ "Gained RCA post-NAFTA",
    had_ca & !gained_ca ~ "Lost RCA post-NAFTA",
    had_ca & gained_ca ~ "Maintained RCA",
    !had_ca & !gained_ca ~ "Never had RCA"
  )) |>
  count(transition) |>
  rename(
    `Transition Status` = transition,
    `Number of Products` = n
  )
```

```

transition_table |>
  gt() |>
  tab_header(
    title = "Comparative Advantage Transitions Around NAFTA (1994)",
    subtitle = "Mexico's Agricultural Sector - HS Codes 01-53"
  ) |>
  tab_spanner(
    label = "Pre vs. Post-NAFTA CA Status",
    columns = everything()
  ) |>
  cols_align(align = "left", columns = `Transition Status`) |>
  cols_align(align = "center", columns = `Number of Products`) |>
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_column_labels()
  ) |>
  tab_style(
    style = cell_fill(color = "#f0f0f0"),
    locations = cells_body(rows = seq(1, nrow(transition_table), 2))
  ) |>
  tab_source_note(
    source_note = "Source: UN Comtrade. RCA calculated using Balassa (1965) index. CA defined"
  ) |>
  opt_table_font(font = "Times New Roman") |> # standard for economics papers
  opt_stylize(style = 1, color = "gray")

```

Miscellaneous and Extra Code

```

##GENERALIZED RCA ANALYSIS

## Generalized RCA function for any product code
#calculate_rca_for_product <- function(product_code) {

  # build file paths for product
  # mx_prod_path <- paste0("mx_xports_prod_i/mx_xports_", product_code, ".csv")
  # world_prod_path <- paste0("world_xports_prod_i/world_xports_", product_code, ".csv")

```

Comparative Advantage Transitions Around NAFTA (1994)

Mexico's Agricultural Sector — HS Codes 01–53] Comparative Advantage Transitions Around NAFTA (1994)

	Pre vs. Post-NAFTA CA Status
	Transition Status
Mexico's Agricultural Sector — HS Codes 01–53	Gained RCA post-NAFTA
	Lost RCA post-NAFTA
	Maintained RCA
	Never had RCA

Source: UN Comtrade. RCA calculated using Balassa (1965) index. CA defined as RCA > 1.

```
# load and read the data
#mx_xports_prod <- read_csv(mx_prod_path, show_col_types = FALSE)
# world_xports_prod <- read_csv(world_prod_path, show_col_types = FALSE)

# load world totals for time frame
#mx_xports_all <- read_csv("mx_total_xports_world/mx_xports_all.csv", show_col_types = FALSE)
#world_xports_all <- read_csv("world_total_xports/world_xports_all.csv", show_col_types = FALSE)

# clean and arrange using necessary_variables function
# mx_xports_prod <- necessary_variables(mx_xports_prod)
# world_xports_prod <- necessary_variables(world_xports_prod)
# mx_xports_all <- necessary_variables(mx_xports_all)
# world_xports_all <- necessary_variables(world_xports_all)

# calculate RCA
# rca_df <- mx_xports_prod %>%
#   inner_join(mx_xports_all, by = "period", suffix = c("_prod", "_total")) %>%
#   inner_join(world_xports_prod, by = "period") %>%
#   inner_join(world_xports_all, by = "period", suffix = c("_world_prod", "_world_total")) %>%
#   mutate(
#     RCA = (fobvalue_prod / fobvalue_total) /
#       (fobvalue_world_prod / fobvalue_world_total)
#   ) %>%
#   select(period, RCA) %>%
#   mutate(product_code = product_code)
```

```
# return(rca_df)
#}
```

run RCA all sectors

```
rca_all <- purrr::map_dfr(sector_codes, process_sector)
```

```
## keep only first two digits
```

```
rca_all <- rca_all |>
  mutate(
    cmdCode = as.character(cmdCode),
    sector = substr(cmdCode, 1, 2)
  )
```

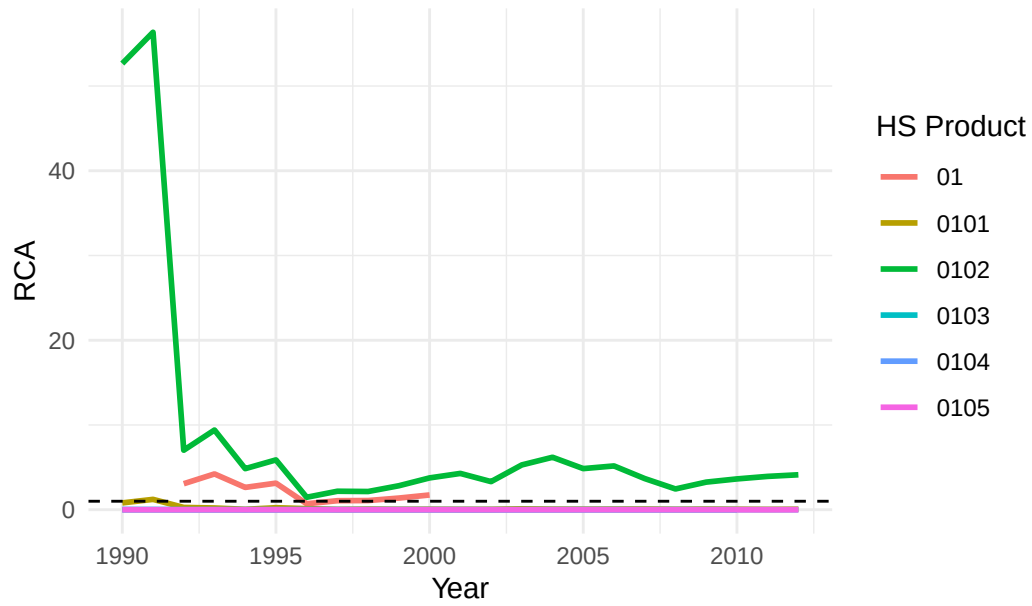
```
sectors <- unique(rca_all$sector) ## creatrs a list for all elements in sector_code MAKE SUR
```

```
for (s in sectors) { ## for all elements s in list sectors,
```

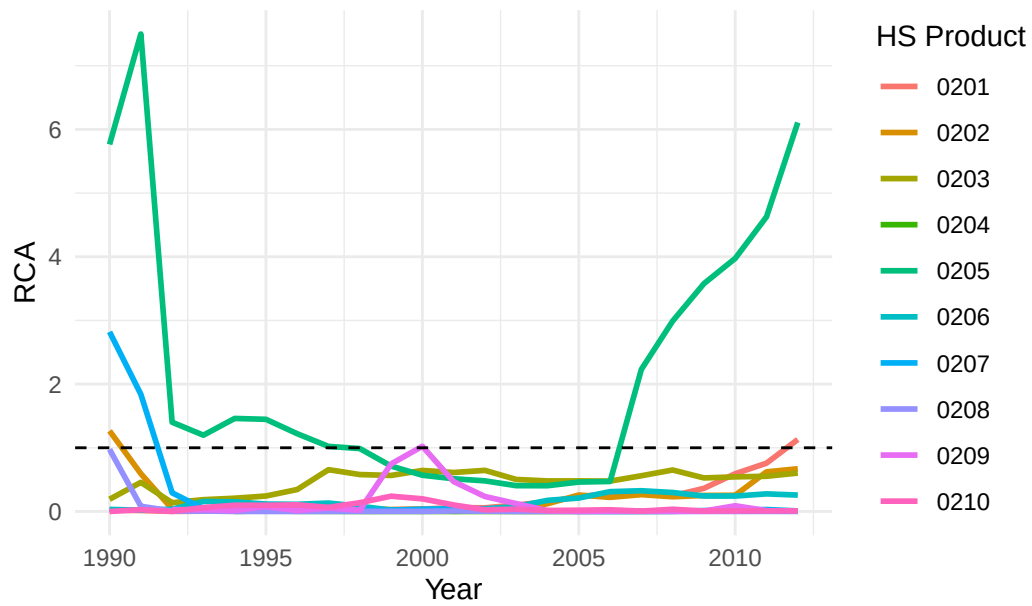
```
  allplots <- ggplot(
    rca_all |> filter(sector == s), ## filter for HS code loops
    aes(x = period, y = RCA, color = cmdCode, group = cmdCode)
  ) +
  geom_line(size = 1) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(
    title = paste("RCA for Sector", s),
    x = "Year",
    y = "RCA",
    color = "HS Product"
  ) +
  theme_minimal()

  print(allplots)
}
```

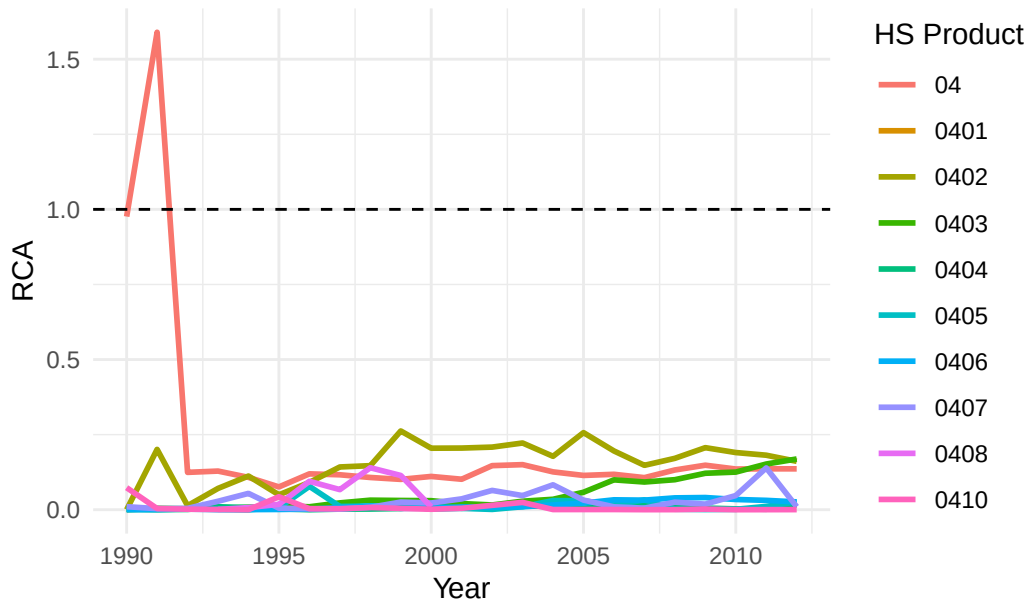
RCA for Sector 01



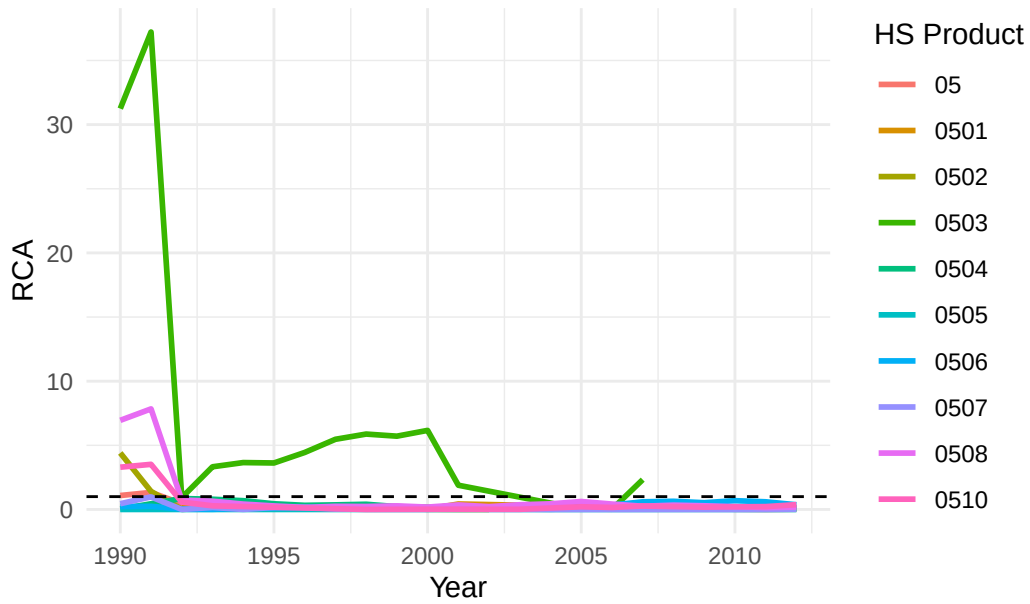
RCA for Sector 02



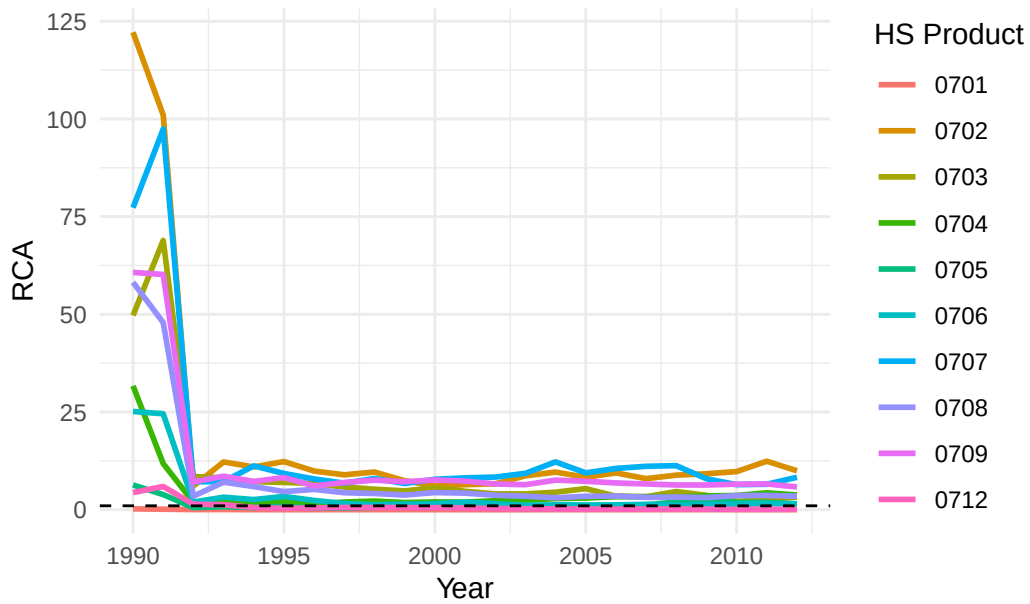
RCA for Sector 04



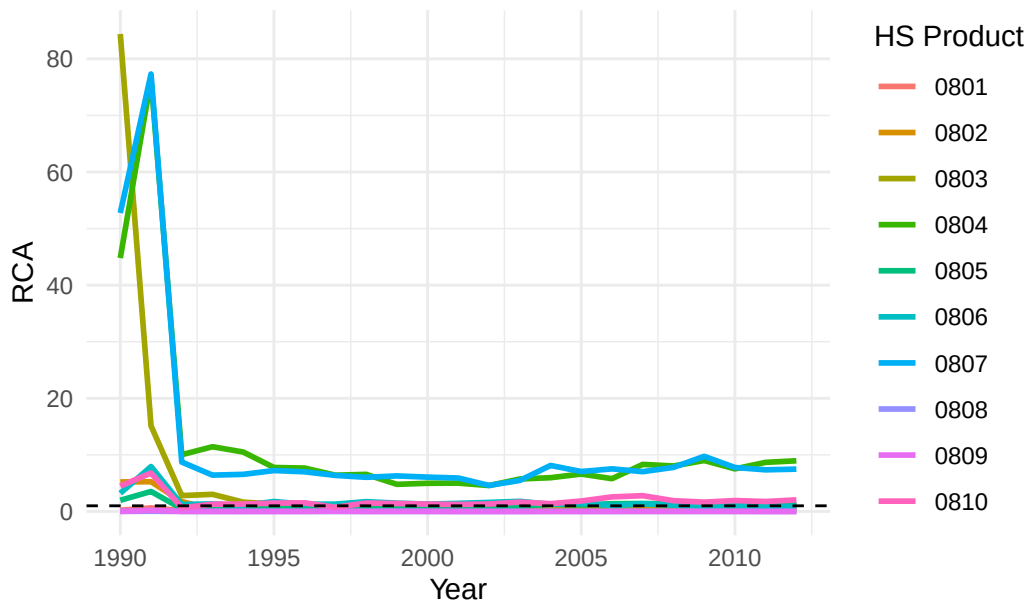
RCA for Sector 05

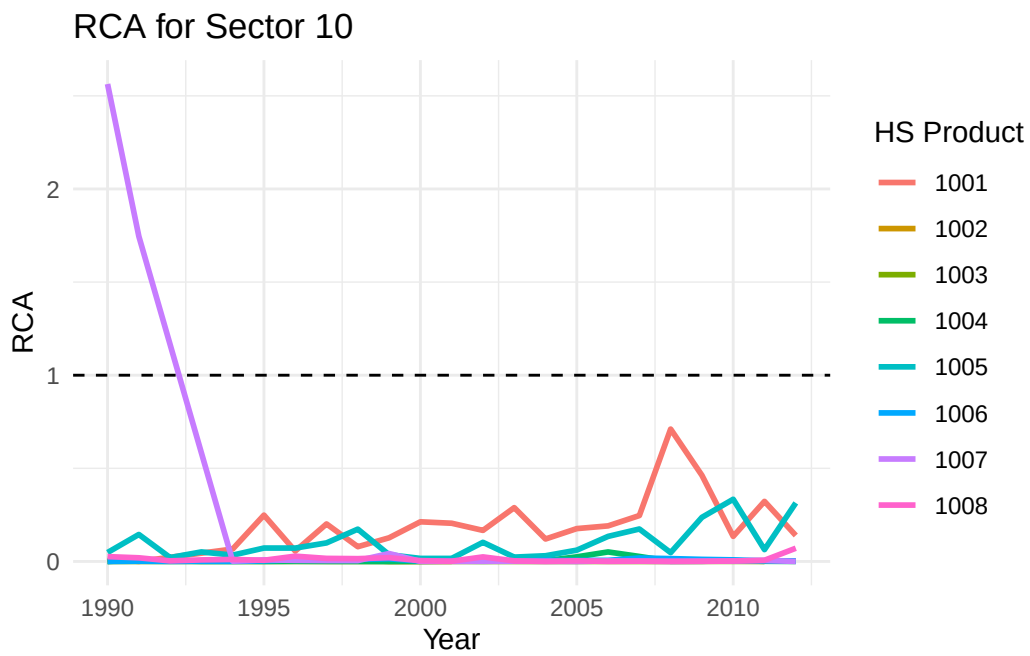
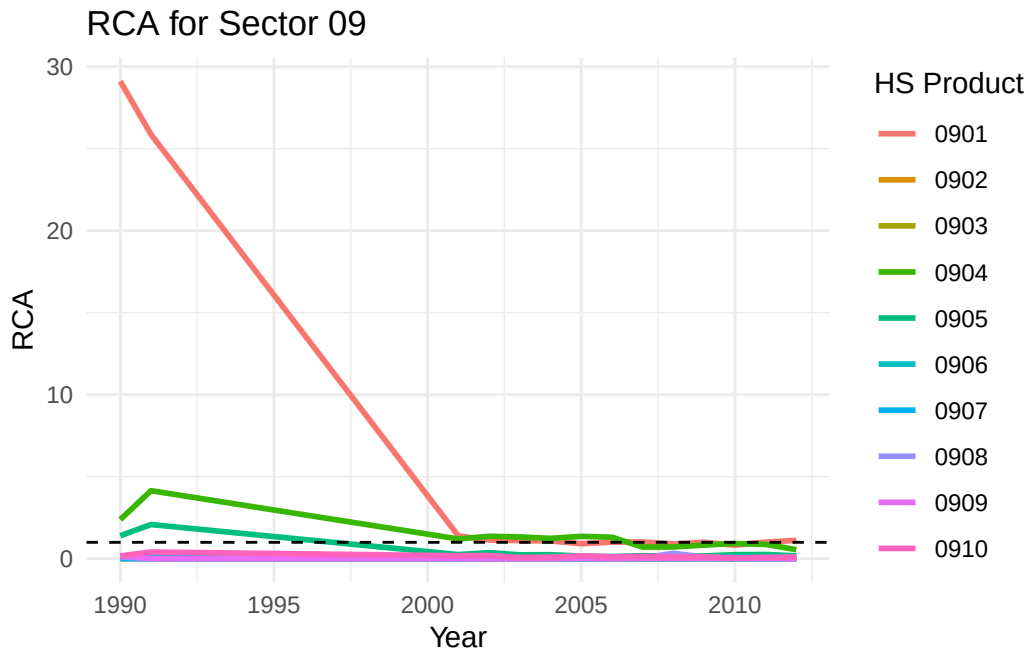


RCA for Sector 07

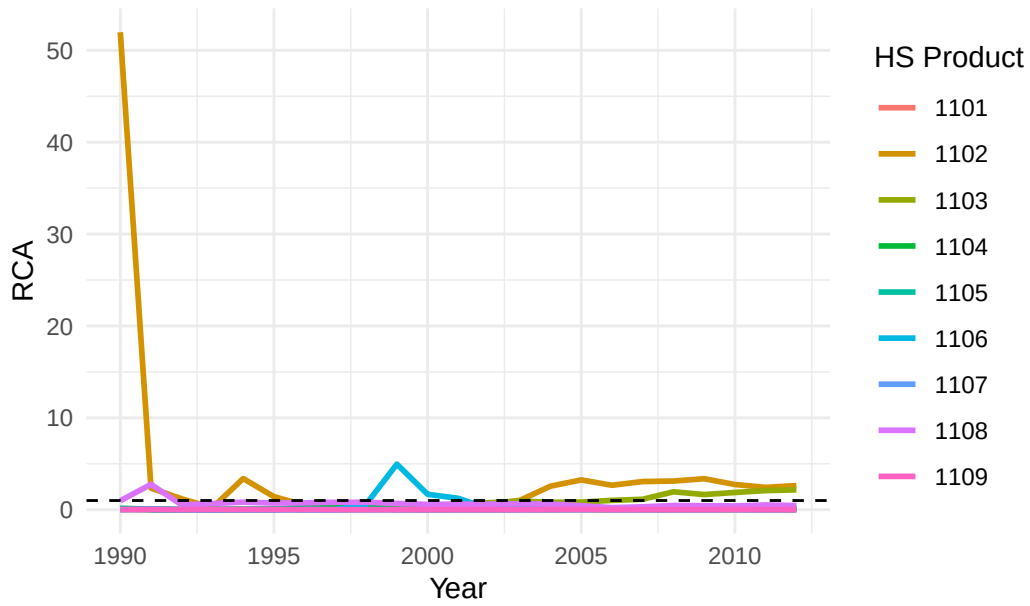


RCA for Sector 08

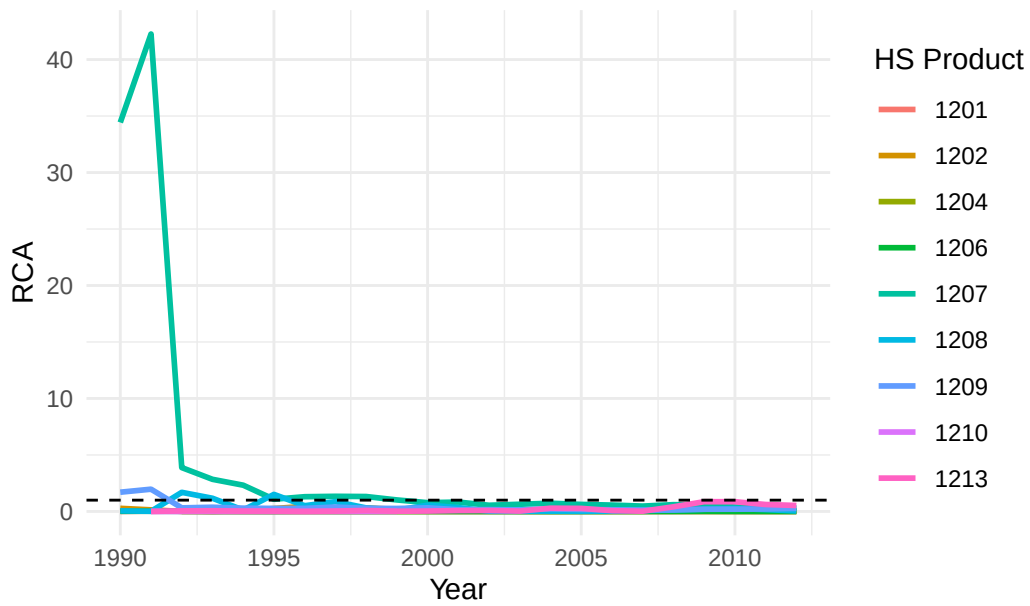


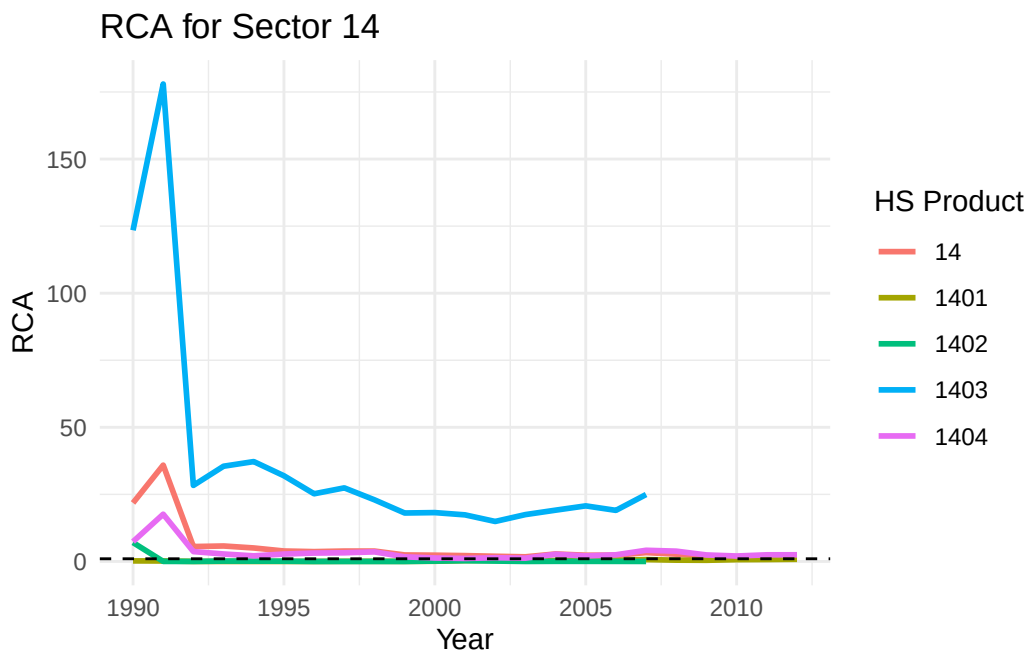
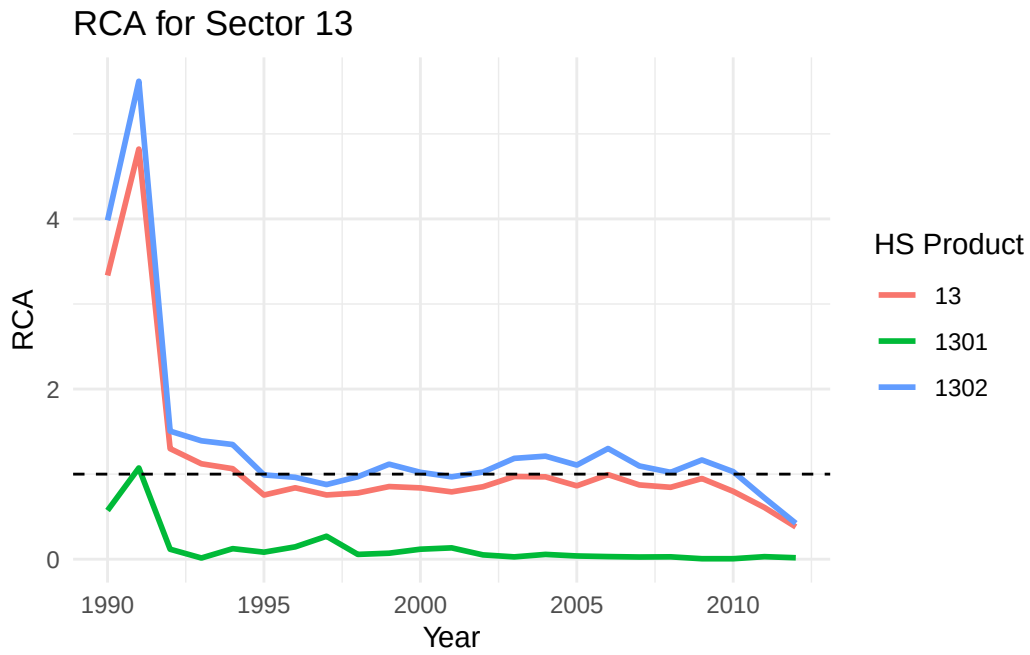


RCA for Sector 11

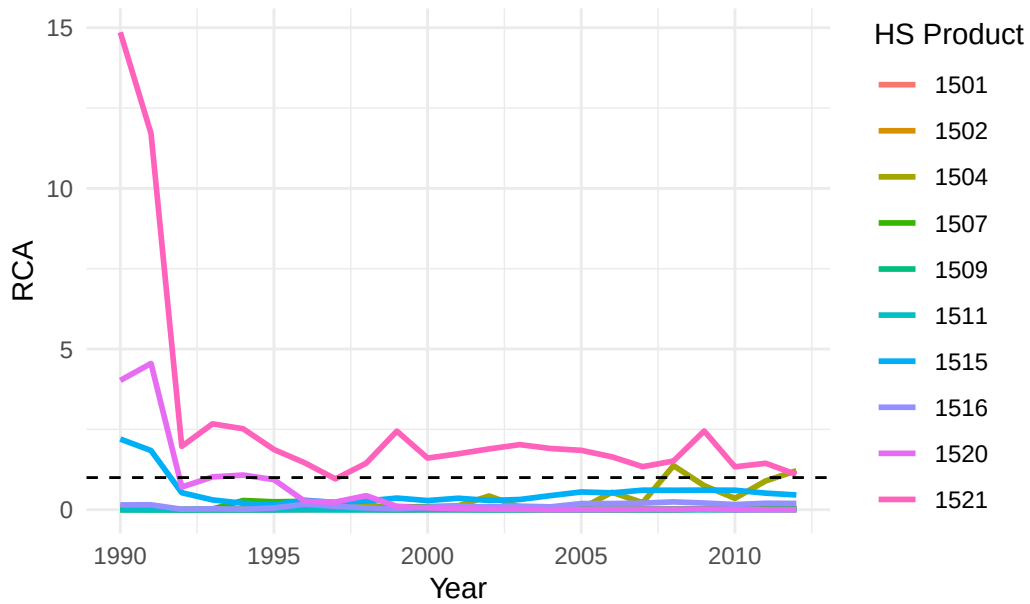


RCA for Sector 12

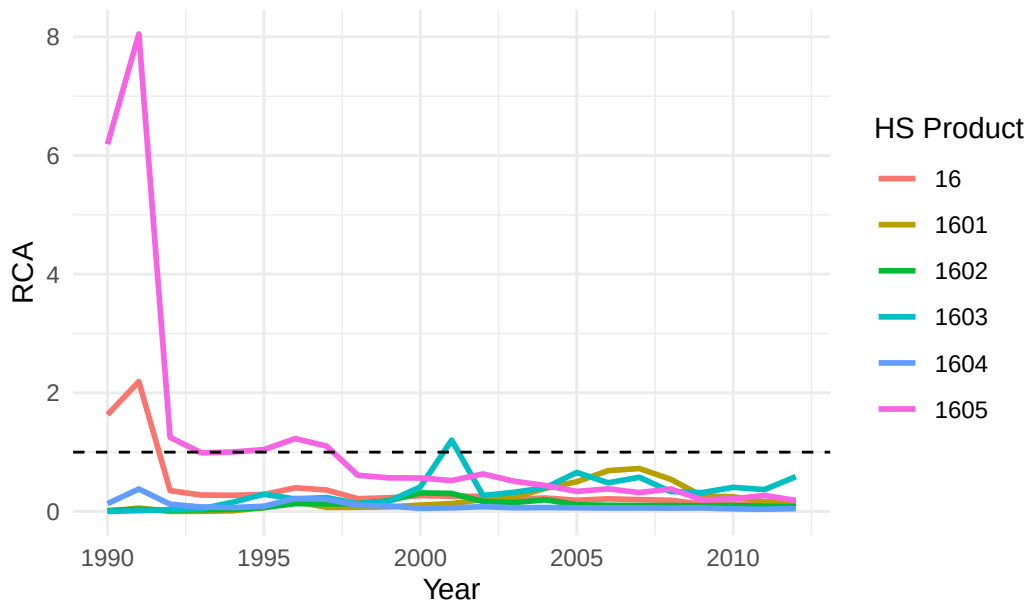


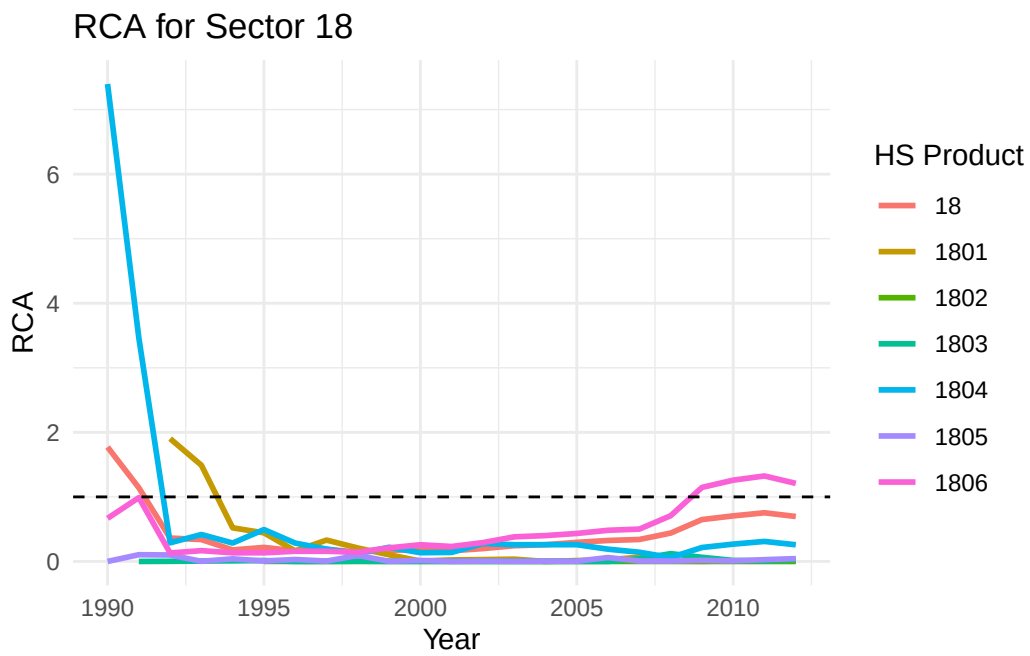
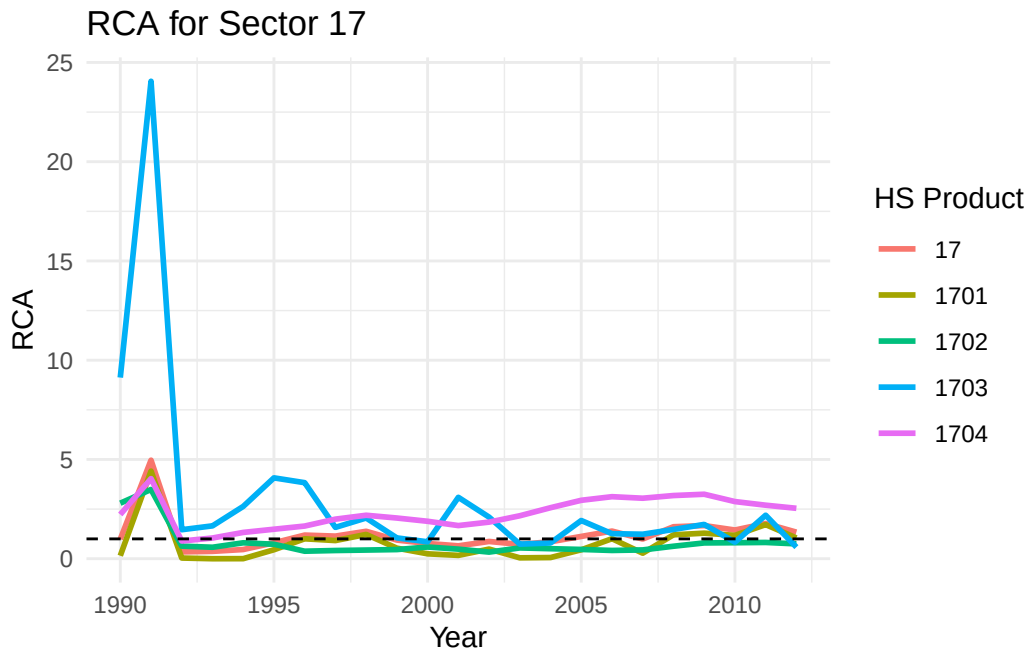


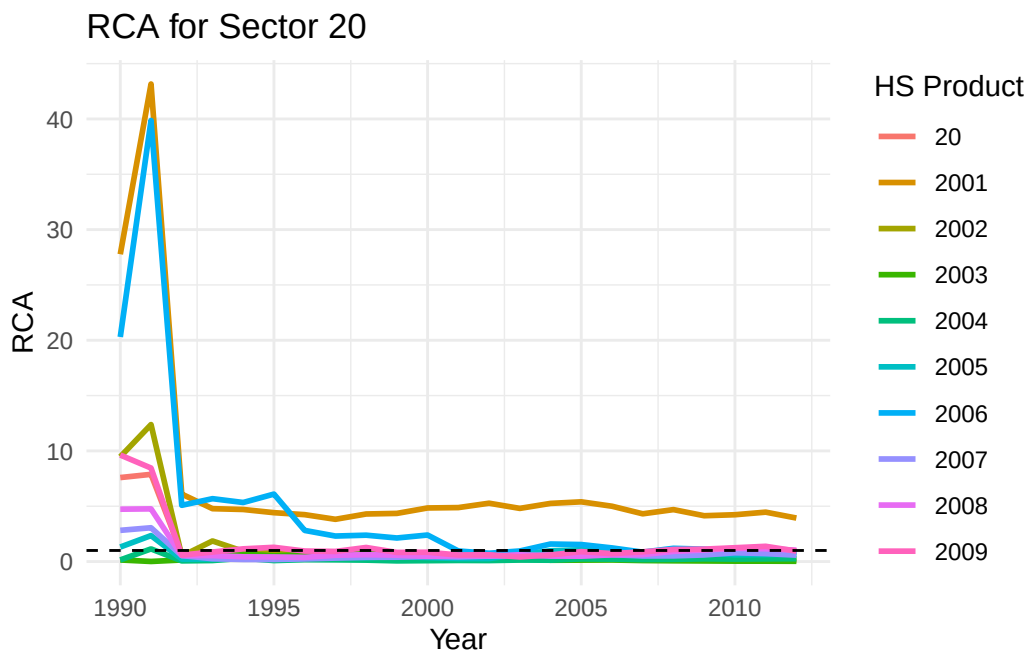
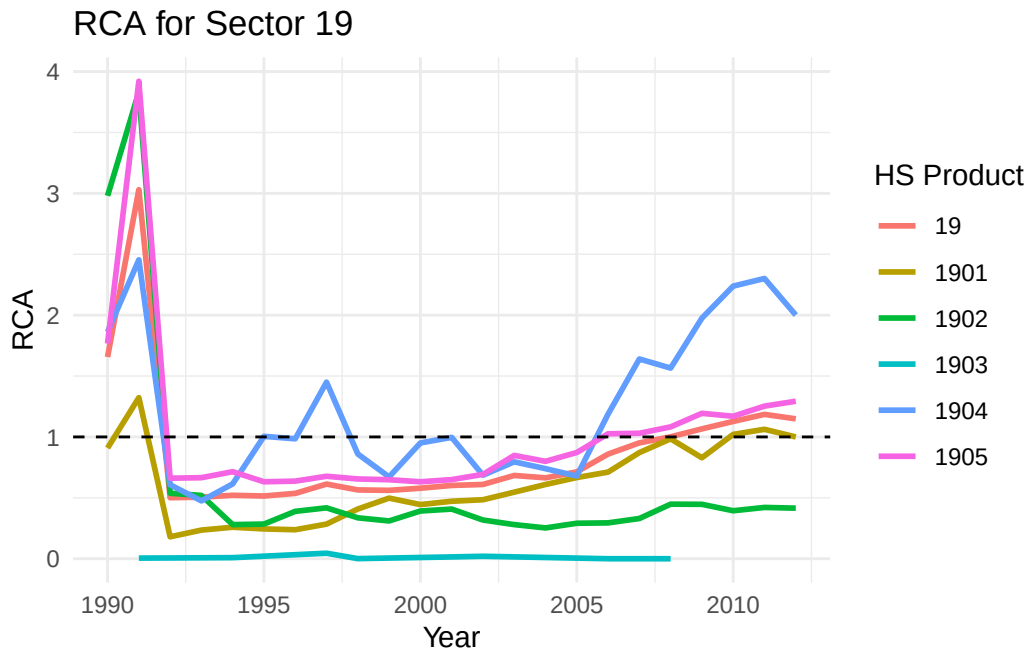
RCA for Sector 15



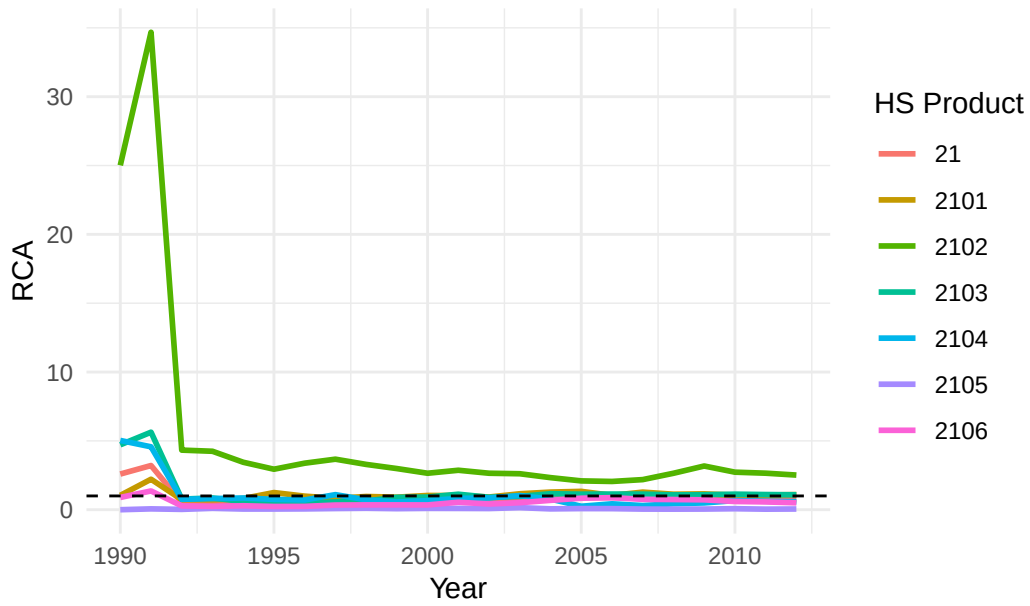
RCA for Sector 16



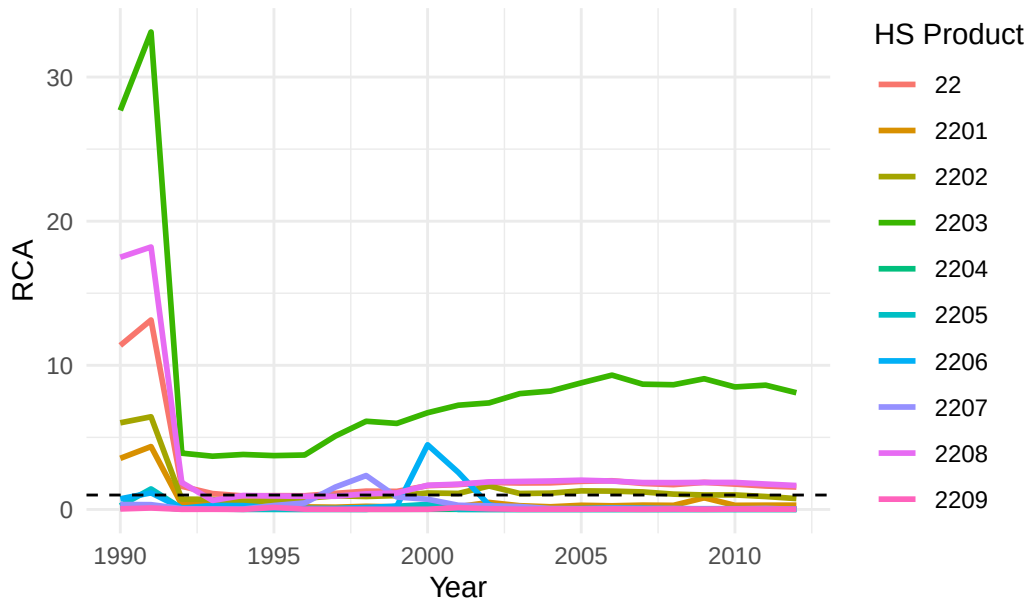




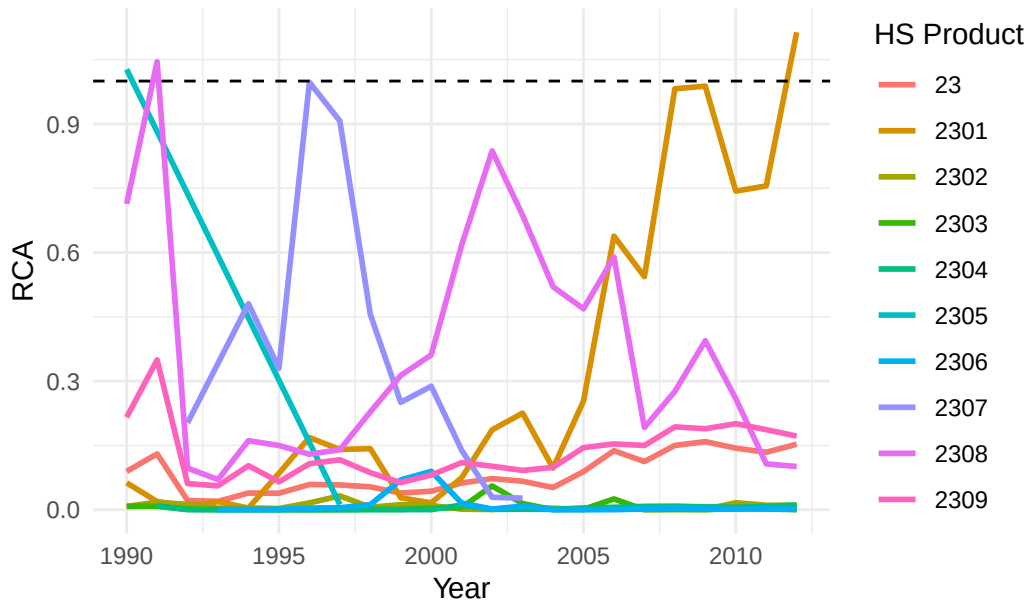
RCA for Sector 21



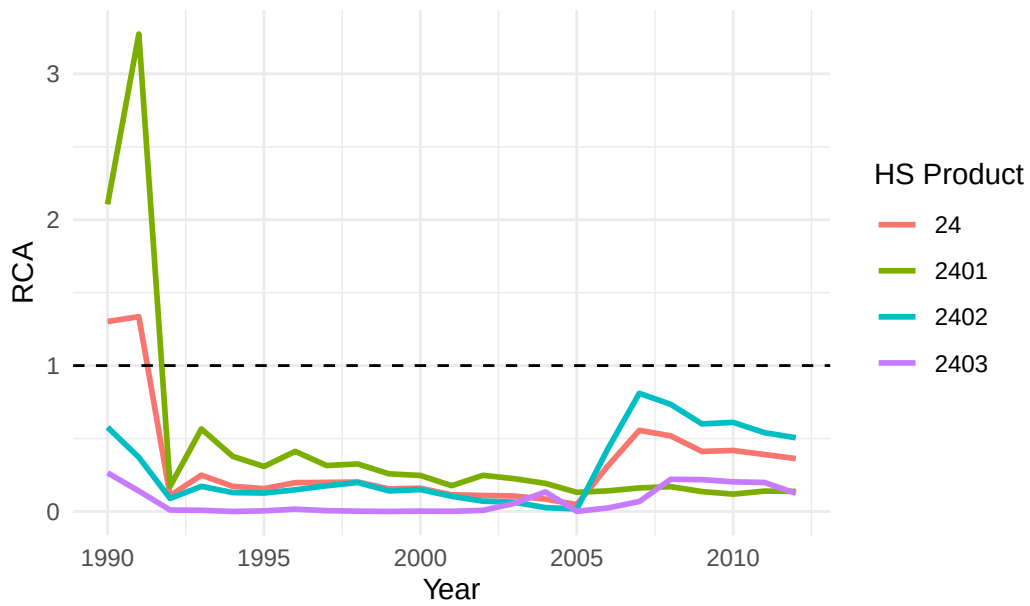
RCA for Sector 22

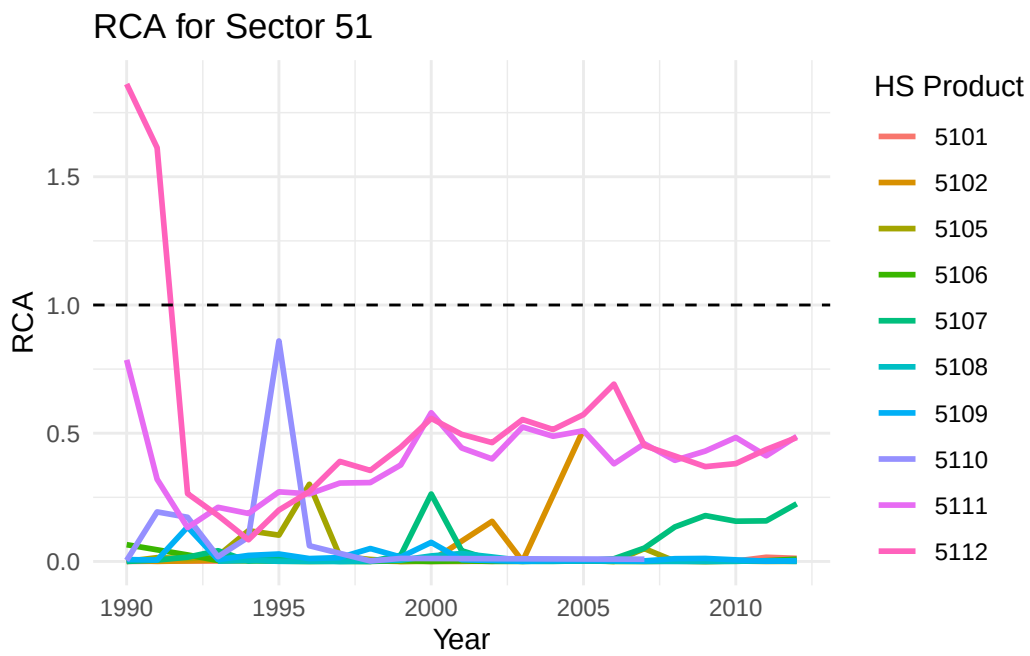
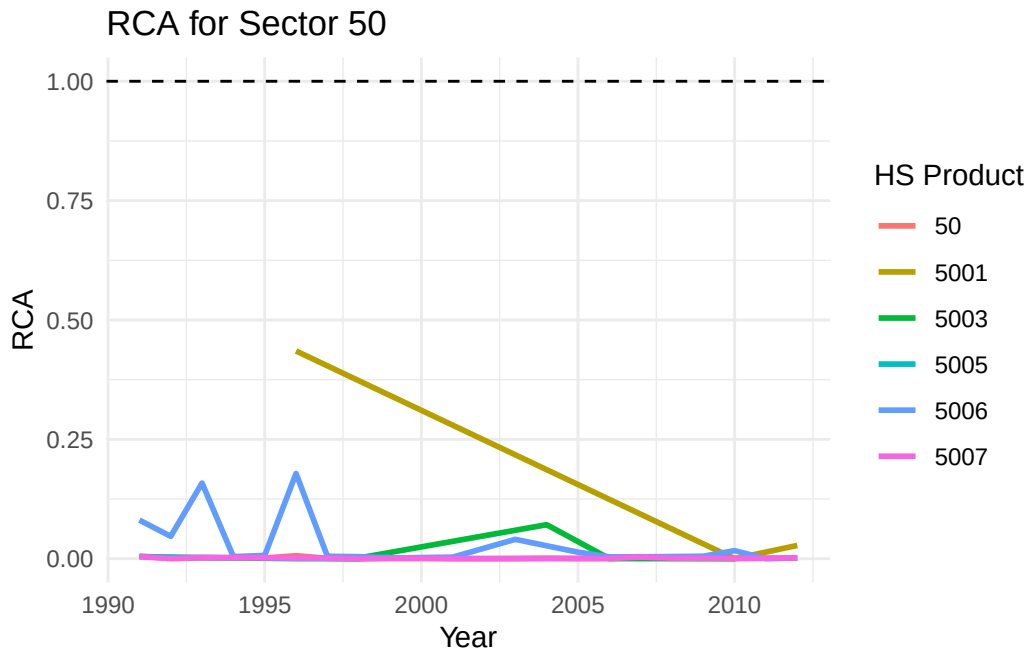


RCA for Sector 23

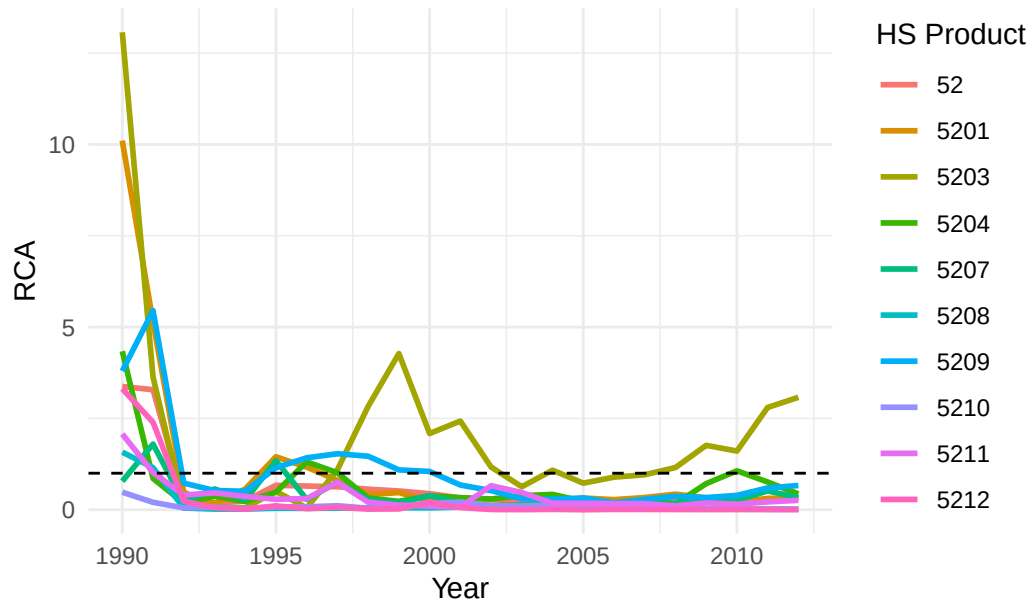


RCA for Sector 24





RCA for Sector 52



RCA for Sector 53

